



# Generic diagonalizability, structural functional observability and output controllability<sup>☆</sup>

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## ABSTRACT

This paper investigates the structural functional observability (SFO) and structural output controllability (SOC) of a class of systems and explores the associated minimal sensor and actuator placement problems. The verification of SOC and the corresponding sensor and actuator placement problems, i.e., the problems of determining the minimum number of outputs and inputs required to achieve SFO and SOC, respectively, are yet open for general systems. This motivates our focus on a large class of systems enabling polynomial-time solutions. In this line, we first define and characterize generically diagonalizable systems, referring to structured systems for which almost all realizations of the state matrices are diagonalizable. We then develop computationally efficient criteria for SFO and SOC within the context of generically diagonalizable systems. Our work expands the class of systems amenable to polynomial-time SOC verification. Thanks to the simplicity of the obtained criteria, we derive closed-form solutions for determining the minimal sensor placement to achieve SFO and the minimal actuator deployment to achieve SOC in such systems, along with efficient weighted maximum matching-based and weighted maximum flow-based algorithms. For more general systems to achieve SFO, we establish an upper bound on the number of required sensors by identifying a non-decreasing property of SFO with respect to a specific class of edge additions, which is proven to be optimal under certain conditions.

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## 1. Introduction

State observation is a fundamental issue within the field of control theory, as it forms a crucial prerequisite for a wide range of practical applications, including observer-based feedback control (Chen, 1984), fault diagnosis (Chen & Patton, 2012), and attack detection and identification (Pasqualetti et al., 2013). The problem of designing a full-state observer has garnered substantial attention since the pioneering work of Luenberger (1966). However, in many real-world scenarios, it is neither practical nor desirable to estimate the entire state variables of a high-dimensional system. Instead, the estimation of specific subsets

or linear combinations of state variables of particular interest is more applicable. A parallel scenario arises in control of large-scale systems/networks, where it is often impractical to fully control the whole states/nodes, but the control of a prescribed subset or combination of states/nodes may suffice (Gao et al., 2014; Kreindler & Sarachik, 1964; Monshizadeh et al., 2015; Van Waarde et al., 2017). The problem of designing functional observers to estimate linear combinations of state variables was also initiated by Luenberger (1971). Since then, significant advancements have been made in functional observer designs for various types of systems (Darouach, 2000, 2012; Fernando et al., 2010b; Jennings et al., 2011; Rotella & Zambettakis, 2015).

The concept of output controllability, introduced by Kreindler and Sarachik (1964) in the 1960s, encompasses the ability to manipulate system's output arbitrarily. By contrast, it was very recently that the concept of functional observability was proposed by Fernando et al. (2010b). Functional observability represents a fundamental condition for the existence of an observer with arbitrary poles to estimate a linear function of states based on system inputs and outputs. Note that there is no direct duality between output controllability and functional observability (Iudice et al., 2019). Various algebraic criteria for functional observability have

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been found (Darouach & Fernando, 2025; Jennings et al., 2011; Rotella & Zambettakis, 2015; Zhang et al., 2025b). These algebraic criteria involve either the system's eigenspace decomposition or observability matrices, emphasizing the necessity of precise parameter values for the system matrices.

Recently, Montanari et al. (2022) have expanded the notion of functional observability to structured systems, introducing the concept of structural functional observability (SFO). Similar to structural controllability (Lin, 1974), SFO relies exclusively on the zero–nonzero structure of system matrices, effectively mitigating the influence of parameter uncertainty and rounding-off errors. This distinctive feature of SFO facilitates the development of scalable combinatorial algorithms for both verifying functional observability and optimizing sensor placement (Montanari et al., 2022). However, it is crucial to highlight that the criterion proposed in Montanari et al. (2022) is based on a PBH-like rank condition for functional observability, a condition acknowledged as insufficient for general systems recently in Darouach and Fernando (2025) and Zhang et al. (2025b). Indeed, the diagonalizability of state matrices significantly impacts the criteria for functional observability in numerical systems, as recognized in Zhang et al. (2025b). Specifically, for diagonalizable systems, a PBH-like criterion exists, extending the classical PBH criterion. This phenomenon can be attributed to the fact that eigenvectors of diagonalizable matrices can span the entire space, unlike non-diagonalizable matrices. In the case of non-diagonalizable systems, criteria must depend on either high-degree products of system matrices (similar to the observability matrix) or the observability canonical decomposition (Darouach & Fernando, 2025). The complete criteria for SFO in general systems, as outlined in Zhang et al. (2025b), are inevitably intricate and require a global understanding of the system structure, such as the generic dimension of unobservable subspaces. This complexity poses a significant challenge to the associated sensor placement problems (Zhang et al., 2025b).

The study of structural output controllability (SOC), on the other hand, has a relatively long history (Czeizler et al., 2018; Gao et al., 2014; Li et al., 2021; Monshizadeh et al., 2015; Murota & Poljak, 1990; Van Waarde et al., 2017; Zhang et al., 2025a). Despite this, it remains open whether SOC can be verified in polynomial time by deterministic algorithms (Murota & Poljak, 1990). However, affirmative answers have been found for certain special classes of systems. For instance, Gao et al. (2014) developed the  $k$ -walk theory for structural target controllability of directed tree networks. Li et al. (2021) provided graph-theoretic characterizations for structural target controllability of undirected networks with symmetric parameters. Recently, Commault et al. (2019) proposed a notion termed (structural) functional output controllability, referring to the ability to produce *any* smooth output trajectory in  $\mathbb{R}^p$  ( $p$  is the dimension of system outputs) by designing system inputs. This notion is equivalent to the right-invertibility of system input–output transfer functions (Commault et al., 2019), which is sufficient for output controllability but not necessary, thus different from the output controllability discussed in this paper. Another research line is strong structural output controllability, which requires *all* system realizations to be output controllable (Monshizadeh et al., 2015). Based on the so-called zero forcing sets, sufficient conditions for the strong structural output (or target) controllability of several classes of structured systems have been given in Monshizadeh et al. (2015), Van Waarde et al. (2017), and Shali et al. (2021).

In this paper, we focus on the SFO and SOC, along with the associated minimal sensor and actuator placement problems, i.e., the problems of determining the minimum number of outputs (sensors) and inputs (actuators) required to achieve SFO and SOC, respectively, within a class of systems with generically

diagonalizable state matrices (termed generically diagonalizable systems). We begin by providing explicit characterizations for generically diagonalizable matrices, referring to structured matrices for which almost all realizations are diagonalizable. We then demonstrate that within the domain of generically diagonalizable systems, the aforementioned verification and optimization problems can be efficiently addressed in polynomial time. We also extend our analysis to encompass general cases. Our main contributions are as follows:

- Firstly, we define generically diagonalizable matrices (along with structurally diagonalizable graphs) and provide their graph-theoretic characterizations. In particular, we establish that a graph is structurally diagonalizable if and only if each subgraph induced by every subset of strongly-connected components (SCCs) of this graph is so. We also propose a weighted maximum matching based algorithm to identify generic diagonalizability.
- Secondly, we develop simplified criteria for SFO in generically diagonalizable systems. These criteria are significantly simpler compared to those for general systems given in Zhang et al. (2025b), as they no longer necessitate the system global information, such as the generic dimension of unobservable subspaces, which is beneficial for solving the subsequent minimal sensor placement problem. Notably, our criteria affirm the applicability of the criterion in Montanari et al. (2022) to the class of generically diagonalizable systems introduced in this paper, despite its derivation from a different SFO definition.
- Thirdly, we identify a class of systems for which the SOC can be verified in polynomial time and highlight that generically diagonalizable systems fall into it.
- Lastly, leveraging the established criteria, we present a closed-form solution and a weighted maximum matching based algorithm for the minimal sensor placement problem for achieving SFO in generically diagonalizable systems. For more general systems, we identify a non-decreasing property of SFO with respect to (w.r.t.) a specific class of edge additions and propose two algorithms to obtain an upper bound, proven optimal under certain circumstances. We also propose a weighted maximum flow algorithm to determine the minimal actuators required for SOC in generically diagonalizable systems. Remarkably, for general systems, both of these problems remain open.

The study of generically diagonalizable matrices holds independent interest, impacting criteria related to various system properties, such as Lyapunov stability of modes with zero real parts (Chen, 1984) and the controllability of networked high-order systems (Xue & Roy, 2021; Zhang et al., 2021). Furthermore, diagonalizable systems find frequent application in practical scenarios, including systems with symmetric state matrices, non-zero random diagonal entries or simple spectra (Tao & Vu, 2017), and distributed estimation (Park & Martins, 2017).

This paper is organized as follows. Section 2 presents some basic preliminaries. Section 3 reviews the concepts of SFO and SOC. Characterizations of generically diagonalizable matrices are provided in Section 4. Computationally efficient criteria for SFO and SOC of generically diagonalizable systems are given respectively in Sections 5 and 6. Section 7 is devoted to the associated minimal sensor and actuator placement problems. Section 8 concludes this paper.

**Notations:** Define the set  $[n] \triangleq \{1, 2, \dots, n\}$  for any integer  $n \geq 1$ . Denote the set of non-negative real numbers by  $\mathbb{R}_{\geq 0}$ . The composite matrix stacked by  $X_1, \dots, X_n$  is denoted as  $\text{col}\{X_i\}_{i=1}^n$ , where  $X_i$  represents the  $i$ th row block. When  $n$  is small, this

matrix can also be denoted by  $[X_1; \dots; X_n]$ . The identity matrix of dimension  $n$  is denoted as  $I_n$ , and the subscript  $n$  can be omitted if inferred from the context. For an  $m \times n$  matrix  $M$  and two sets  $S_1 \subseteq [m]$  and  $S_2 \subseteq [n]$ , the sub-matrix of  $M$  formed by rows indexed by  $S_1$  and columns indexed by  $S_2$  is denoted as  $M(S_1, S_2)$ . When  $S_1 = [m]$  ( $S_2 = [n]$ ),  $M(S_1, S_2)$  is also denoted as  $M(:, S_2)$  ( $M(S_1, :)$ ). The number of rows of a matrix  $M$  is denoted by  $\text{row}(M)$ . The rank of  $M$  is given by  $\text{rank } M$ .

## 2. Preliminaries

This section introduces preliminaries in graph theory and structured system theory (Dion et al., 2003; Ramos et al., 2022).

**Graph-theoretic terminologies:** A directed graph (digraph) is represented by  $\mathcal{G} = (V, E)$ , where  $V$  is the vertex set and  $E \subseteq V \times V$  is the edge set. A subgraph  $\mathcal{G}_s = (V_s, E_s)$  of  $\mathcal{G}$  is a graph such that  $V_s \subseteq V$  and  $E_s \subseteq E$ . It is called a subgraph induced by  $V_s$  if  $E_s = (V_s \times V_s) \cap E$ . We say that  $\mathcal{G}_s = (V_s, E_s)$  spans  $\mathcal{G} = (V, E)$  if  $V_s = V$ , and that  $\mathcal{G}_s$  covers a subset  $V'_s \subseteq V$  if  $V'_s \subseteq V_s$ . A path  $P$  from vertex  $i_1$  to vertex  $i_k$  in  $\mathcal{G}$  is a sequence of edges  $(i_1, i_2)$ ,  $(i_2, i_3)$ , ...,  $(i_{k-1}, i_k)$  with  $(i_j, i_{j+1}) \in E$ ,  $j = 1, \dots, k-1$ . Denote such a path by  $P = (i_1, i_2, \dots, i_k)$ . If there is a path from  $i_1$  to  $i_k$ , we say  $i_k$  is *reachable* from  $i_1$ . A path with no repeated vertices is referred to as a simple path. The length of a path is the number of edges it contains. A vertex can be seen as a path with zero length. A bipartite graph  $\mathcal{B} = (V, E)$  is a graph where the vertex set  $V$  can be partitioned into two disjoint sets  $V_L$  and  $V_R$  (called parts of  $V$ ) such that  $E \subseteq V_R \times V_L$ . In this case,  $\mathcal{B}$  is also denoted by  $\mathcal{B} = (V_L, V_R, E)$ .<sup>1</sup> A matching of  $\mathcal{B}$  is a set of edges  $\mathcal{M} \subseteq E$  in which no two edges share a common end vertex. The size of  $\mathcal{M}$ , denoted as  $|\mathcal{M}|$ , is the number of edges in  $\mathcal{M}$ . A vertex  $v$  is considered *left matched* (resp. *right matched*) by  $\mathcal{M}$  if  $v \in V_L$  (resp.  $v \in V_R$ ) is incident to an edge in  $\mathcal{M}$ . A *perfect matching* is a matching that left matches  $V_L$  and right matches  $V_R$ . A maximum matching of  $\mathcal{B}$  refers to a matching with the largest size among all possible matchings of  $\mathcal{B}$ . If each edge  $e \in E$  is assigned a non-negative cost, the weight of a matching is the sum of the costs of its edges. The weighted maximum matching problem seeks the maximum (or minimum) weight of a maximum matching among all maximum matchings. This problem can be solved in polynomial time (Ahuja et al., 1993). A *strongly connected component* (SCC) of a directed graph  $\mathcal{G} = (V, E)$  is a subgraph  $\mathcal{G}_s$  of  $\mathcal{G}$  that satisfies the following conditions: for any pair of vertices in  $\mathcal{G}_s$ , there exists a path from either vertex to the other, and no additional vertices can be included in  $\mathcal{G}_s$  without violating the previous property.

**Structured system theory:** Consider a linear-time invariant (LTI) system described by

$$\dot{x}(t) = Ax(t) + Bu(t), \quad (1a)$$

$$y(t) = Cx(t), \quad (1b)$$

where  $x(t) \in \mathbb{R}^n$  is the state vector,  $u(t) \in \mathbb{R}^m$  is the input vector, and  $y(t) \in \mathbb{R}^p$  is the output vector.

A structured matrix is a matrix whose entries are either fixed zero or free parameters that can take arbitrary real values independently (i.e., no parameter dependence exists among those parameters). We denote the set of  $n_1 \times n_2$  structured matrices by  $\{0, *\}^{n_1 \times n_2}$ , where  $*$  represents the free parameters, and 0 the fixed zero entries. For a structured matrix  $M \in \{0, *\}^{n_1 \times n_2}$ , let  $n_{\bar{M}}$  denote the number of  $*$  entries in  $\bar{M}$ . A realization is obtained by assigning some specific values to the  $*$  entries. We use the vector gathering values for  $*$  entries  $\theta \in \mathbb{R}^{n_{\bar{M}}}$  to denote a realization

of  $\bar{M}$  (with some abuse of notation, since they are of one-to-one correspondence). For a subset  $\mathbb{V} \subseteq \mathbb{R}^{n_{\bar{M}}}$ ,  $M \in \mathbb{V}$  means that  $M$  is a realization of  $\bar{M}$  corresponding to some  $\theta \in \mathbb{V}$ .

An algebraic variety in  $\mathbb{R}^d$  is the set of real-valued solutions of a system of polynomial equations in  $d$  variables. A proper variety in  $\mathbb{R}^d$  is an algebraic variety that is not the entire space  $\mathbb{R}^d$  (i.e., it is defined by at least one non-zero polynomial equation). The *generic rank* of a structured matrix  $\bar{M}$ , given by  $\text{grank } \bar{M}$ , is the maximum rank the realizations of  $\bar{M}$  can achieve as a function of its free parameters. Furthermore, let the entries  $P_{ij}$  of a matrix  $P$  be polynomials in  $d$  free parameters (for example,  $P$  can be the product of a sequence of structured matrices in  $d$  free parameters). Its *generic rank*  $\text{grank } P$  is the maximum rank this matrix can achieve as a function of the  $d$  free parameters in  $P$ . Here, the generic rank also equals the rank that this matrix can achieve for almost all choices of parameter values (i.e., all except for some proper variety) in the parameter space  $\mathbb{R}^d$  (Murota (2009, page 38)). For a structured matrix  $\bar{M} \in \{0, *\}^{n_1 \times n_2}$ , the bipartite graph associated with  $\bar{M}$ , given by  $\mathcal{B}(\bar{M})$ , is defined as  $(X_L, X_R, E_{RL})$ , where  $X_L, X_R$  correspond to respectively the rows and columns of  $\bar{M}$ , and  $E_{RL} = \{(i, j) : \bar{M}_{ji} \neq 0, i \in X_R, j \in X_L\}$ . It follows from Murota (2009, Props. 2.1.12 and 2.2.25) that  $\text{grank } \bar{M}$  equals the size of a maximum matching of  $\mathcal{B}(\bar{M})$ .

Consider a triple  $(\bar{A}, \bar{B}, \bar{C})$  with  $\bar{A} \in \{0, *\}^{n \times n}$ ,  $\bar{B} \in \{0, *\}^{n \times m}$ , and  $\bar{C} \in \{0, *\}^{p \times n}$ . We define the *digraph*  $\mathcal{G}(\bar{A}, \bar{B}, \bar{C}) = (X \cup U \cup Y, E_{UX} \cup E_{XX} \cup E_{XY})$  as follows: the state vertices  $X = \{x_1, \dots, x_n\}$ , the input vertices  $U = \{u_1, \dots, u_m\}$ , the output vertices  $Y = \{y_1, \dots, y_p\}$ , and the edges  $E_{XX} = \{(x_i, x_j) : \bar{A}_{ji} \neq 0\}$ ,  $E_{UX} = \{(u_i, x_j) : \bar{B}_{ji} \neq 0\}$ , and  $E_{XY} = \{(x_i, y_j) : \bar{C}_{ji} \neq 0\}$ . Let  $\mathcal{G}(\bar{A})$ ,  $\mathcal{G}(\bar{A}, \bar{B})$ , and  $\mathcal{G}(\bar{A}, \bar{C})$  be the subgraphs of  $\mathcal{G}(\bar{A}, \bar{B}, \bar{C})$  induced by  $X, X \cup U$ , and  $X \cup Y$ , respectively. Define a bipartite graph  $\mathcal{B}(\bar{A}) = (X, X, E_{XX})$ , where with some abuse of notations, we allow two vertices in different parts to have the same label. In this way, for a matching  $\mathcal{M}$  in  $\mathcal{B}(\bar{A})$ ,  $(X, \mathcal{M})$  is a subgraph of  $\mathcal{G}(\bar{A})$  with vertex set  $X$  and edge set  $\mathcal{M} \subseteq E_{XX}$ . A state vertex  $x_i \in X$  is considered *output-reachable* (resp. *input-reachable*) if  $x_i$  is reachable to some output vertex  $y_j \in Y$  in  $\mathcal{G}(\bar{A}, \bar{C})$  (resp. reachable from some input vertex  $u_j \in U$  in  $\mathcal{G}(\bar{A}, \bar{B})$ ). A state vertex  $x_i \in X$  has a self-loop if  $(x_i, x_i) \in E_{XX}$ . An *output stem* is a simple path from a state vertex to an output vertex in  $\mathcal{G}(\bar{A}, \bar{C})$ . Similarly, an *input stem* is a simple path from an input vertex to a state vertex in  $\mathcal{G}(\bar{A}, \bar{B})$ . An *output cactus configuration* of  $\mathcal{G}(\bar{A}, \bar{C})$  is a subgraph consisting of a collection of vertex-disjoint output stems and cycles, where each vertex in the cycles is output-reachable in  $\mathcal{G}(\bar{A}, \bar{C})$  (notice that the paths making each cycle output-reachable are not included in a cactus configuration). Similarly, we can define *input cactus configuration* by changing “output stems” to “input stems” in  $\mathcal{G}(\bar{A}, \bar{B})$ . The size of an input (output) cactus configuration is the number of state vertices it covers.

A pair  $(\bar{A}, \bar{C})$  (resp.  $(\bar{A}, \bar{B})$ ) is said to be *structurally observable* (*structurally controllable*) if there is a realization  $(A, C)$  of  $(\bar{A}, \bar{C})$  that is observable (resp. a realization  $(A, B)$  of  $(\bar{A}, \bar{B})$  that is controllable) (Lin, 1974). It is known that the pair  $(\bar{A}, \bar{C})$  is structurally observable (resp.  $(\bar{A}, \bar{B})$  is structurally controllable) if and only if the whole state vertex set  $X$  is covered by an output cactus configuration in  $\mathcal{G}(\bar{A}, \bar{C})$  (resp. input cactus configuration in  $\mathcal{G}(\bar{A}, \bar{B})$ ). See Commault et al. (2008, Theo 1) and Dion et al. (2003, Theo 1) for some equivalent conditions for structural observability.

## 3. SFO and SOC

Consider an LTI system described by

$$\dot{x}(t) = Ax(t) + Bu(t), \quad (2a)$$

$$y(t) = Cx(t), \quad (2b)$$

$$z(t) = Fx(t), \quad (2c)$$

<sup>1</sup> Although edges in a bipartite graph are unoriented, in this paper it is convenient to consider that they have a direction from  $V_R$  to  $V_L$ .



where  $x(t) \in \mathbb{R}^n$  is the state vector,  $u(t) \in \mathbb{R}^m$  is the input vector,  $y(t) \in \mathbb{R}^p$  is the output vector, and  $z(t) \in \mathbb{R}^r$  is the functional of states to be estimated. Accordingly, matrices  $A \in \mathbb{R}^{n \times n}$ ,  $B \in \mathbb{R}^{n \times m}$ ,  $C \in \mathbb{R}^{p \times n}$ , and  $F \in \mathbb{R}^{r \times n}$ . Throughout this paper,  $O(A, C) = \text{col}\{C, CA, \dots, CA^{n-1}\}$  denotes the observability matrix of the pair  $(A, C)$ , and  $Q(A, B) = [B, AB, \dots, A^{n-1}B]$  denotes the controllability matrix of the pair  $(A, B)$ . 5

**Definition 1** (Ogata & Yang, 2002). System (2) (or the triple  $(A, B, C)$ ) is output controllable if, for any initial output  $y_0 \in \mathbb{R}^p$  and any final output  $y_f \in \mathbb{R}^p$ , there exist a finite time  $T$  and control input  $u(t) : [0, T] \rightarrow \mathbb{R}^m$  such that  $y(0) = y_0$  and  $y(T) = y_f$ .

**Definition 2** (Fernando et al., 2010a). System (2) (or the triple  $(A, C, F)$ ) is said to be functionally observable if, for any initial state  $x(0)$  and input  $u(t)$ , there exists a finite time  $T$  such that the value of  $Fx(0)$  can be uniquely determined from the output  $y(t)$  and input  $u(t)$ ,  $0 \leq t \leq T$ .

As mentioned earlier, the functional observability of  $(A, C, F)$  is exactly the existence condition for a functional observer with arbitrary poles to estimate the linear function  $z(t) = Fx(t)$  based on system input  $u(t)$  and output  $y(t)$  (Fernando et al., 2010b). It is easy to see that when  $F = I_n$ , functional observability collapses to the conventional observability (Chen, 1984). The following lemma presents a PBH-like criterion for the functional observability of a class of systems with diagonalizable state matrices  $A$ , which are called *diagonalizable systems* throughout. It is worth mentioning that this condition is necessary, but generically insufficient, for non-diagonalizable systems (Zhang et al., 2025b).

**Lemma 1** (Darouach & Fernando, 2025; Zhang et al., 2025b). Suppose  $A$  is diagonalizable. The triple  $(A, C, F)$  is functionally observable if and only if

$$\text{rank} \begin{bmatrix} A - \lambda I \\ C \\ F \end{bmatrix} = \text{rank} \begin{bmatrix} A - \lambda I \\ C \end{bmatrix}, \forall \lambda \in \mathbb{C}. \quad (3)$$

Let  $\bar{A} \in \{0, *\}^{n \times n}$ ,  $\bar{B} \in \{0, *\}^{n \times m}$ ,  $\bar{C} \in \{0, *\}^{p \times n}$ , and  $\bar{F} \in \{0, *\}^{r \times n}$  be structured matrices specifying the zero-nonzero structure of  $A, B, C$ , and  $F$ , respectively. That is,  $M_{ij} = 0$  implies  $M_{ij} = 0$  for  $M = A, B, C$  or  $F$ . The triple  $(\bar{A}, \bar{B}, \bar{C})$  is SOC if there is a realization of  $(\bar{A}, \bar{B}, \bar{C})$  that is output controllable (Czeizler et al., 2018; Murota & Poljak, 1990). By contrast, SFO is defined differently, shown as follows.

**Definition 3** (Zhang et al., 2025b). The triple  $(\bar{A}, \bar{C}, \bar{F})$  is SFO if there is a proper variety  $\mathbb{V} \subseteq \mathbb{R}^{n_{\bar{A}}} \times \mathbb{R}^{n_{\bar{C}}} \times \mathbb{R}^{n_{\bar{F}}}$  (which is a variety in  $\mathbb{R}^{n_{\bar{A}}+n_{\bar{B}}+n_{\bar{C}}}$ ), such that all realizations  $(A, C, F) \in \mathbb{R}^{n_{\bar{A}}} \times \mathbb{R}^{n_{\bar{C}}} \times \mathbb{R}^{n_{\bar{F}}} \setminus \mathbb{V}$  are functionally observable.

In other words,  $(\bar{A}, \bar{C}, \bar{F})$  is SFO if *almost all* realizations of  $(\bar{A}, \bar{C}, \bar{F})$  are functionally observable. From Zhang et al. (2025b), if  $(\bar{A}, \bar{C}, \bar{F})$  is not SFO, then almost all its realizations are not functionally observable, except for possibly a proper variety in  $\mathbb{R}^{n_{\bar{A}}} \times \mathbb{R}^{n_{\bar{C}}} \times \mathbb{R}^{n_{\bar{F}}}$ . It is worth noting that Montanari et al. (2022) has defined SFO in the same way as structural observability, i.e., as the property of enabling a functionally observable realization. As demonstrated in Zhang et al. (2025b) and also illustrated in the following example, there is a key distinction between SFO and structural observability. In structural observability, the existence of an observable realization implies that almost all realizations are observable. However, the existence of a functionally observable realization cannot guarantee that almost all realizations are functionally observable.

**Example 1.** Consider a structured system with

$$\bar{A} = \begin{bmatrix} 0 & 0 & 0 & * \\ 0 & 0 & 0 & * \\ 0 & 0 & 0 & * \\ 0 & 0 & 0 & * \end{bmatrix}, \quad \bar{C} = \begin{bmatrix} * & * & * & 0 \\ * & * & * & 0 \\ 0 & 0 & 0 & * \\ * & 0 & 0 & 0 \end{bmatrix}, \quad \bar{F} = \begin{bmatrix} * & 0 & 0 & 0 \end{bmatrix}.$$

There is a realization of  $(\bar{A}, \bar{C}, \bar{F})$  as

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}.$$

It can be verified that  $A$  is diagonalizable and  $(A, C, F)$  is functionally observable via Lemma 1. However, it turns out that  $\text{grank}[\bar{A}; \bar{C}; \bar{F}] = 4 \neq \text{grank}[\bar{A}; \bar{C}] = 3$ . This means for almost all realizations  $(A, C, F) \in \mathbb{R}^{n_{\bar{A}}} \times \mathbb{R}^{n_{\bar{C}}} \times \mathbb{R}^{n_{\bar{F}}}$ ,  $\text{rank}[A; C; F] = 4 \neq \text{rank}[A; C] = 3$ , leading to the functional unobservability of  $(A, C, F)$  via Lemma 1.

**Lemma 2** (Prop 2, Zhang et al., 2025b). The following statements are equivalent:

- (i) The triple  $(\bar{A}, \bar{C}, \bar{F})$  is SFO.
- (ii)  $\text{grank}([O(\bar{A}, \bar{C}); \bar{F}]) = \text{grank } O(\bar{A}, \bar{C})$ .
- (iii)  $\text{grank}([O(\bar{A}, \bar{C}); O(\bar{A}, \bar{F})]) = \text{grank } O(\bar{A}, \bar{C})$ .

It is known that  $(\bar{A}, \bar{B}, \bar{C})$  is SOC if and only if (Murota & Poljak, 1990)

$$\text{grank } \bar{C}Q(\bar{A}, \bar{B}) = p.$$

The primary goal of this paper is to characterize SFO, SOC, and investigate the associated minimal sensor and actuator placement problems within the class of diagonalizable systems, demonstrating their polynomial time solvability. Specially, the minimal sensor and actuator placement problems aim to minimize the number of sensors (rows of  $\bar{C}$ ) and actuators (columns of  $\bar{B}$ ) required to achieve SFO and SOC, respectively, which will be formally defined in Section 7. To this end, we begin by defining ‘diagonalizable’ structured matrices/systems and characterizing this class of matrices, referred to as generically diagonalizable matrices. Then, we develop computationally efficient criteria for SFO and SOC in generically diagonalizable systems. Finally, we provide closed-form solutions for the aforementioned minimal sensor and actuator placement problems in such systems, as well as an upper bound for general systems.

#### 4. Characterizations of generically diagonalizable matrices

In this section, we define and characterize generically diagonalizable matrices. We also relate generic diagonalizability to SCCs and provide an efficient algorithm to verify generic diagonalizability. Our characterizations can be used to identify generically diagonalizable systems studied in the subsequent sections.

**Definition 4** (Generic Diagonalizability). A structured matrix  $\bar{A} \in \{0, *\}^{n \times n}$  is called generically diagonalizable if almost all realizations of  $\bar{A}$  are diagonalizable, i.e., there exists a proper variety  $\mathbb{V} \subseteq \mathbb{R}^{n_{\bar{A}}}$  such that all realizations  $A \in \mathbb{R}^{n_{\bar{A}}} \setminus \mathbb{V}$  are diagonalizable. System (2) is said to be generically diagonalizable if  $\bar{A}$  is generically diagonalizable.

Note that the existence of a diagonalizable realization does not imply generic diagonalizability (for example,  $0_{n \times n}$  is a diagonalizable realization of any  $\bar{A} \in \{0, *\}^{n \times n}$ ).

**Lemma 3** (Hosoe & Matsumoto, 1979). Given a structured pair  $(\bar{A}, \bar{C})$ , there exists a proper variety  $\mathbb{V} \subseteq \mathbb{R}^{n_{\bar{A}}} \times \mathbb{R}^{n_{\bar{C}}}$  such that for all realizations  $(A, C) \in \mathbb{R}^{n_{\bar{A}}} \times \mathbb{R}^{n_{\bar{C}}} \setminus \mathbb{V}$ , (1) every nonzero eigenvalue of  $A$  is simple (i.e., with algebraic multiplicity one), and (2) if all

state vertices in  $\mathcal{G}(\bar{A}, \bar{C})$  are output-reachable, then each nonzero eigenvalue (mode) of  $A$  is observable, i.e.,  $\text{rank}[A - \lambda I_n; C] = n$  for each nonzero eigenvalue  $\lambda$  of  $A$ .

**Lemma 4** (Theo A2.1, Reinschke, 1988; Lem 3, Pequeto et al., 2016). Given  $\bar{A} \in \{0, *\}^{n \times n}$ , let  $\mathcal{M}$  be any maximum matching of  $\mathcal{B}(\bar{A})$ . Then,  $(X, \mathcal{M})$  is a union of disjoint cycles and paths, and the number of paths equals  $n - |\mathcal{M}|$  (an isolated vertex is regarded as a path with length zero), recalling that  $(X, \mathcal{M})$  is a digraph with vertex set  $X$  and edge set  $\mathcal{M} \subseteq E_{XX}$  that spans  $\mathcal{G}(\bar{A})$ .

**Theorem 1.** Given a structured matrix  $\bar{A} \in \{0, *\}^{n \times n}$ , the following statements are equivalent:

- (a)  $\bar{A}$  is generically diagonalizable.
- (b)  $\text{grank} \bar{A} = v(\bar{A})$ , where  $v(\bar{A})$  is the maximum number of vertices that are covered by a collection of (vertex) disjoint cycles in  $\mathcal{G}(\bar{A})$ .
- (c) There is a maximum matching  $\mathcal{M}$  of  $\mathcal{B}(\bar{A})$  such that the digraph  $\mathcal{G}(\mathcal{M}) \doteq (X, \mathcal{M})$  is a union of disjoint cycles and isolated vertices.

**Proof** (a).  $\Leftrightarrow$  (b): Let  $A$  be a realization of  $\bar{A}$ . Expand  $\det(\lambda I - A)$  as

$$\det(\lambda I - A) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_n,$$

where, by the definition of determinant, the coefficients

$$a_k = (-1)^k \sum_{J=\{j_1, \dots, j_k\} \subseteq [n], j_1 < \dots < j_k} \det A(J, J), \forall k \in [n].$$

Let  $\mu(\bar{A}) = \max\{|J| : \text{grank} \bar{A}(J, J) = |J|, J \subseteq [n]\}$ . Using Lemma 4 on  $\mathcal{B}(\bar{A}(J, J))$ ,  $J \subseteq [n]$ , we have  $\mu(\bar{A}) = v(\bar{A})$ . It then follows that  $a_k = 0$  if  $k > v(\bar{A})$ . Therefore,  $\det(\lambda I - A)$  can be expressed as

$$\det(\lambda I - A) = \lambda^{n-v(\bar{A})} \phi_A(\lambda),$$

where  $\phi_A(\lambda) = \lambda^{v(\bar{A})} + a_1 \lambda^{v(\bar{A})-1} + \dots + a_{v(\bar{A})}$ . It has been proven in Hosoe and Matsumoto (1979, Lem 2) that for almost all realizations  $A$  of  $\bar{A}$ , all  $v(\bar{A})$  roots of  $\phi_A(\lambda)$  are mutually distinct and nonzero (see Lemma 3). Consequently, to make  $A$  diagonalizable, it suffices to let the algebraic multiplicity of the zero eigenvalue of  $A$ , which is  $n - v(\bar{A})$ , equal its geometric multiplicity, which is  $n - \text{grank} \bar{A}$ , for almost all realizations  $A$  of  $\bar{A}$ . Therefore, condition (b) is sufficient for the generic diagonalizability of  $\bar{A}$ . On the other hand, if condition (b) is not satisfied, then the zero eigenvalue has its geometric multiplicity not equaling its algebraic multiplicity for almost all realizations  $A$  of  $\bar{A}$ , leading to non-diagonalizability of  $A$ .

(b)  $\Leftrightarrow$  (c): If (c) holds, then  $\text{grank} \bar{A}$  equals exactly the number of vertices that are covered by the union of disjoint cycles. By Lemma 4, no decomposition of  $\mathcal{G}(\bar{A})$  into disjoint cycles and paths can contain disjoint cycles that can cover more vertices than  $\text{grank} \bar{A}$ . As a result,  $\text{grank} \bar{A} = v(\bar{A})$ , which is (b). On the other hand, if (b) holds, let  $\mathcal{C}$  be a union of disjoint cycles that covers the maximum number of vertices. Then,  $\mathcal{C}$  corresponds to a matching  $\mathcal{M}$  of  $\mathcal{B}(\bar{A})$  and no other edge from  $\mathcal{B}(\bar{A})$  can be added to  $\mathcal{M}$  to form a larger matching as  $|\mathcal{M}| = v(\bar{A}) = \text{grank} \bar{A}$ , which is exactly (c).  $\square$

The following corollary is immediate from the proof of Theorem 1.

**Corollary 1.** Given a structured matrix  $\bar{A} \in \{0, *\}^{n \times n}$ , either almost all realizations of  $\bar{A}$  are diagonalizable, or almost all realizations of  $\bar{A}$  are non-diagonalizable.

In view of Corollary 1, if  $\bar{A}$  is not generically diagonalizable, we call it generically non-diagonalizable, reflecting the fact that almost all its realizations are non-diagonalizable. With some abuse of terminology, if  $\bar{A}$  is generically diagonalizable, we say the digraph  $\mathcal{G}(\bar{A})$  is *structurally diagonalizable*. Below, we present some easily verified classes of structurally diagonalizable graphs.

**Corollary 2.** If each vertex of  $\mathcal{G}(\bar{A})$  has a self-loop, then  $\bar{A}$  is generically diagonalizable. If  $\mathcal{G}(\bar{A})$  is acyclic (i.e., without any cycles, including self-loops) and  $\bar{A} \neq 0$ , then  $\bar{A}$  is generically non-diagonalizable.

**Proof.** The first statement is immediate from Theorem 1(c). If there is no cycle in  $\mathcal{G}(\bar{A})$  and  $\bar{A} \neq 0$ , then every maximum matching  $\mathcal{M}$  of  $\mathcal{B}(\bar{A})$  corresponds to  $\mathcal{G}(\mathcal{M})$  with at least one path with nonzero length. According to Theorem 1(c),  $\bar{A}$  is generically non-diagonalizable.  $\square$

**Corollary 3.** If  $\bar{A}$  is structurally symmetric, possibly with nonzero diagonal entries (i.e.,  $\bar{A}_{ij} \neq 0$  implies  $\bar{A}_{ji} \neq 0$ ,  $\forall i, j$ ; equivalently, every edge of  $\mathcal{G}(\bar{A})$  is bidirectional except for the self-loops, or simply put,  $\mathcal{G}(\bar{A})$  is undirected),<sup>2</sup> then  $\bar{A}$  is generically diagonalizable.

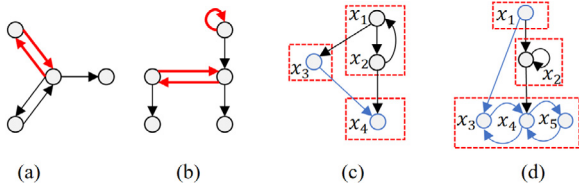
**Proof.** Let  $\mathcal{M}$  be a maximum matching of  $\mathcal{B}(\bar{A})$ . We first show that  $\mathcal{G}(\mathcal{M}) = (X, \mathcal{M})$  is a union of disjoint cycles and paths, where each path has an even length. Suppose, for the sake of contradiction, there is a path  $P = (x_{i_1}, x_{i_2}, \dots, x_{i_k})$  with an odd length in  $\mathcal{G}(\mathcal{M})$ , i.e.,  $k \geq 2$  is even. By the symmetry of  $\bar{A}$ , there exist  $k/2$  cycles in  $\mathcal{G}(\bar{A})$ , consisting of edges  $\{(x_{i_1}, x_{i_2}), (x_{i_2}, x_{i_1}), \dots, \{(x_{i_{k-1}}, x_{i_k}), (x_{i_k}, x_{i_{k-1}})\}$ . The  $k/2$  cycles contain  $k$  edges in total, and are disjoint with the remaining cycles and paths of  $\mathcal{G}(\mathcal{M})$  (except for  $P$ ). This corresponds to a matching of size  $|\mathcal{M}| + 1$  in  $\mathcal{B}(\bar{A})$ , contradicting the assumption that  $\mathcal{M}$  is a maximum matching. For each path with an even length in  $(X, \mathcal{M})$ , say  $P' = (x_{i_1}, x_{i_2}, \dots, x_{i_k})$ , where  $k \geq 3$  is odd, due to the symmetry of  $\bar{A}$ , there are  $(k-1)/2$  cycles in  $\mathcal{G}(\bar{A})$ , consisting of edges  $\{(x_{i_2}, x_{i_3}), (x_{i_3}, x_{i_2}), \dots, \{(x_{i_{k-1}}, x_{i_k}), (x_{i_k}, x_{i_{k-1}})\}$ . In other words,  $P'$  can be covered by a union of disjoint cycles and isolated vertices, and this union contains the same number  $k-1$  of edges as  $P'$ . It is then easy to see that there is a maximum matching  $\mathcal{M}'$  of  $\mathcal{B}(\bar{A})$  such that  $(X, \mathcal{M}')$  is a union of disjoint cycles and isolated vertices. The required result then follows from Theorem 1.  $\square$

It should be noted that the class of structurally diagonalizable graphs is much broader than the two cases mentioned above. For example, Figs. 1(a) and (b) present two structurally diagonalizable digraphs not falling into these cases. Another example is when  $\bar{A}$  has full generic rank, in which case  $\mathcal{G}(\bar{A})$  is called structurally cyclic in the literature (Moothedath et al., 2019). Characterizing the distribution of structurally diagonalizable graphs within the set of all graphs is a subject for our future research.

**Proposition 1** (Generic Diagonalizability & SCC). A structured matrix  $\bar{A} \in \{0, *\}^{n \times n}$  is generically diagonalizable if and only if every subgraph of  $\mathcal{G}(\bar{A})$  induced by the union of vertices of each subset of SCCs of  $\mathcal{G}(\bar{A})$  (i.e., the subgraph induced by  $\bigcup_{i \in S} X_i$  for each subset  $S \subseteq \{1, \dots, r\}$ , where  $r$  is the number of SCCs of  $\mathcal{G}(\bar{A})$ , and  $X_i$  is the vertex set of the  $i$ th SCC) is structurally diagonalizable.

**Proof.** Notice that the complete set  $S = \{1, \dots, r\}$  corresponds to exactly  $\mathcal{G}(\bar{A})$ . The sufficiency follows immediately. For the necessity, since  $\bar{A}$  is generically diagonalizable, by (c) of Theorem 1, there is a maximum matching  $\mathcal{M}$  of  $\mathcal{B}(\bar{A})$  such that  $\mathcal{G}(\mathcal{M})$  is the

<sup>2</sup> This does not mean all realizations of  $\bar{A}$  are symmetric. Instead, the symmetric realizations correspond to a set of zero measure in  $\mathbb{R}^{n^2}$ .



**Fig. 1.** (a) and (b): Examples of structurally diagonalizable graphs. Bold (red) edges correspond to a maximum matching  $\mathcal{M}$ , associated with which  $\mathcal{G}(\mathcal{M}) = (X, \mathcal{M})$  is a union of disjoint cycles and isolated vertices. (c) and (d): Examples of structurally non-diagonalizable graphs. Each SCC is in a box, while the subgraphs in blue are induced by a specific set of SCCs. In (c), the subgraph induced by  $\{x_3, x_4\}$  is acyclic, and thus is structurally non-diagonalizable (Corollary 2). In (d), the subgraph induced by  $\{x_1, x_3, x_4, x_5\}$  is structurally non-diagonalizable as any maximum matching of it cannot correspond to disjoint cycles and isolated vertices. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

union of disjoint cycles and isolated vertices. By the definition of SCC, all edges of a cycle of  $\mathcal{G}(\mathcal{M})$  must belong to the same SCC. As a result, given any  $S \subseteq \{1, \dots, r\}$ , the subgraph of  $\mathcal{G}(\mathcal{M})$  induced by the vertices  $X_S \doteq \bigcup_{i \in S} X_i$ , denoted by  $\mathcal{G}(\mathcal{M}, S)$ , is the union of disjoint cycles and isolated vertices, too. Denote by  $\mathcal{G}(\bar{A}, S)$  the subgraph of  $\mathcal{G}(\bar{A})$  induced by  $X_S$ , and its associated bipartite graph is given by  $\mathcal{B}(\bar{A}, S)$ . It turns out that the set of edges of  $\mathcal{G}(\mathcal{M}, S)$  corresponds to a maximum matching  $\mathcal{M}_S$  of  $\mathcal{B}(\bar{A}, S)$ . Indeed, if there is a matching  $\mathcal{M}'$  of  $\mathcal{B}(\bar{A}, S)$  whose size is bigger than  $|\mathcal{M}_S|$ , then  $\mathcal{M}' \cup \mathcal{M}_{S^c}$  is a matching of  $\mathcal{B}(\bar{A})$  with size bigger than  $|\mathcal{M}|$ , where  $\mathcal{M}_{S^c} \doteq \mathcal{M} \setminus \mathcal{M}_S$ , noting that the end vertices of  $\mathcal{M}_{S^c}$  belong to  $X \setminus X_S$ . This contradicts the fact that  $\mathcal{M}$  is a maximum matching of  $\mathcal{B}(\bar{A})$ . Since  $\mathcal{G}(\mathcal{M}, S)$  consists of disjoint cycles and isolated vertices, by Theorem 1(c),  $\mathcal{G}(\bar{A}, S)$  is structurally diagonalizable.  $\square$

It is known that if  $\mathcal{G}(\bar{A})$  has  $r$  SCCs, then there is a permutation matrix  $P$  such that  $P\bar{A}P^{-1}$  has a block-triangular form, where the  $i$ th diagonal block corresponds to the vertex set  $X_i$  of the  $i$ th SCC,  $i = 1, \dots, r$  (Murota, 2009). From this view, when  $|S| = 1$ , Proposition 1 coincides with the property that for a block-triangular numerical matrix to be diagonalizable, each of its diagonal blocks should be diagonalizable (this can be proved by contradiction). However, when  $|S| > 1$ , a similar property does not hold for a numerical matrix (i.e., the property that the submatrix with rows and columns corresponding to two or more diagonal blocks should be diagonalizable). For instance,

consider the matrix  $A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$ . It turns

out that  $A$  is diagonalizable. However, the sub-matrix consisting of the 1st and 3rd row blocks and column blocks of  $A$  is non-diagonalizable. Nevertheless, Proposition 1 implies that this property holds generically for numerical matrices. This proposition provides a way to verify the generic non-diagonalizability of large-dimensional structured matrices using local information, i.e., from a subset of SCCs of the associated graphs. In case  $\mathcal{G}(\bar{A}, S)$  (the subgraph of  $\mathcal{G}(\bar{A})$  induced by  $\bigcup_{i \in S} X_i$ ) is structurally non-diagonalizable for some  $S \subseteq \{1, \dots, r\}$ , it follows that  $\bar{A}$  is so (see Figs. 1(c) and (d) for illustration). This proposition will play a role in proving a criterion of SOC (Corollary 4) in Section 6.

In what follows, we provide a weighted maximum matching based algorithm to verify generic diagonalizability, which avoids enumerating all maximum matchings of  $\mathcal{B}(\bar{A})$  in verifying condition (c) of Theorem 1.

**Proposition 2.** Given a structured matrix  $\bar{A} \in \{0, *\}^{n \times n}$ , the problem of verifying whether  $\bar{A}$  is generically diagonalizable can be reduced to a weighted maximum matching problem, and hence is solvable in  $O(n^3)$  time.

**Proof.** Define an auxiliary bipartite graph  $\bar{\mathcal{B}}(\bar{A}) = (X, X, E_{XX} \cup E_\lambda)$  associated with  $\bar{A}$ , where  $E_\lambda = \{(x_i, x_i) : \bar{A}_{ii} = 0\}$ . Assign the cost  $c(e) : E_{XX} \cup E_\lambda \rightarrow \mathbb{N}$  as follows:

$$c(e) = \begin{cases} 1, & \text{if } e \in E_\lambda \\ 0, & \text{if } e \in E_{XX} \end{cases}$$

We will show that  $\bar{A}$  is generically diagonalizable if and only if the minimum weight of a maximum matching of  $\bar{\mathcal{B}}(\bar{A})$ , given by  $\text{MWMM}(\bar{A})$ , equals  $n - \text{grank } \bar{A}$ . By condition (b) of Theorem 1, it suffices to prove that  $v(\bar{A}) = n - \text{MWMM}(\bar{A})$ . Observe that the size of a maximum matching of  $\bar{\mathcal{B}}(\bar{A})$  is  $n$  since  $\{(x_i, x_i) : i = 1, \dots, n\}$  is such a maximum matching. Notice that the disjoint cycles covering  $v(\bar{A})$  vertices in  $\mathcal{G}(\bar{A})$  together with  $n - v(\bar{A})$  edges in  $E_\lambda$  form a maximum matching of  $\bar{\mathcal{B}}(\bar{A})$  with weight  $n - v(\bar{A})$ . By the assignment of edge costs, this means  $\text{MWMM}(\bar{A}) \leq n - v(\bar{A})$ . Now assume  $\text{MWMM}(\bar{A}) < n - v(\bar{A})$ . Let  $\mathcal{M}$  be a maximum matching of  $\bar{\mathcal{B}}(\bar{A})$  with  $\text{MWMM}(\bar{A})$  edges belonging to  $E_\lambda$ , and denote these edges by  $\mathcal{M}' \subseteq E_\lambda$ . Let  $X_1 \subseteq X$  be the set of vertices left matched associated with  $\mathcal{M}'$  and  $X_2 \doteq X \setminus X_1$ . Further, let  $\mathcal{G}(\bar{A}, X_2)$  be the subgraph of  $\mathcal{G}(\bar{A})$  induced by  $X_2$  and  $\mathcal{B}(\bar{A}, X_2)$  be the bipartite graph associated with  $\mathcal{G}(\bar{A}, X_2)$ . Observe that all edges in  $\mathcal{M} \setminus \mathcal{M}'$  have end vertices in  $X_2$ . Consequently,  $\mathcal{M} \setminus \mathcal{M}'$  is a perfect matching of  $\mathcal{B}(\bar{A}, X_2)$ , which corresponds to a union of disjoint cycles that spans  $\mathcal{G}(\bar{A}, X_2)$  according to Lemma 4. By the definition of  $v(\bar{A})$ , this leads to  $v(\bar{A}) \geq |X_2| = n - |X_1|$ , causing a contradiction to the assumption  $v(\bar{A}) < n - |X_1|$ . Therefore,  $\text{MWMM}(\bar{A}) = n - v(\bar{A})$ . The proof is then accomplished with the fact that the weighted maximum matching problem on a bipartite graph with  $n$  vertices can be solved in  $O(n^3)$  time (Ahuja et al., 1993).  $\square$

## 5. SFO criteria for generically diagonalizable systems

In this section, we present criteria for the SFO of generically diagonalizable systems.

Some notations and concepts are introduced first. A *dilation* in  $\mathcal{G}(\bar{A}, \bar{C})$  is a set of state vertices  $X_s \subseteq X$  satisfying  $|\mathcal{N}^o(X_s)| < |X_s|$ . Here,  $\mathcal{N}^o(X_s) \doteq \{v \in X \cup Y : (x, v) \in E_{XX} \cup E_{XY}, x \in X_s\}$  denotes the set of out-neighbors of  $X_s$ . A *minimal dilation* is a dilation such that any of its proper subsets is not a dilation. Given  $\bar{F} \in \{0, *\}^{r \times n}$ , let  $X_F = \{x_i \in X : \bar{F}(:, i) \neq 0\}$  be the set of state vertices corresponding to nonzero columns of  $\bar{F}$ , i.e., states that contribute directly to  $z(t)$ , which are called *functional states*. Let  $\bar{e}_i \in \{0, *\}^{1 \times n}$  be such that the  $i$ th entry of  $\bar{e}_i$  is  $*$  and the rest are zero. Define  $\bar{I}_{X_F} \doteq \text{col}\{\bar{e}_i : x_i \in X_F\}$ , and  $\bar{I}_n \doteq \text{col}\{\bar{e}_i : x_i \in X\}$ .

Given a numerical triple  $(A, C, F)$ , let  $\lambda$  be an eigenvalue of  $A$ . If the algebraic multiplicity of  $\lambda$  equals its geometric multiplicity, then  $\lambda$  is said to be *modal functionally observable* if condition (3) holds for this particular  $\lambda$  (Zhang et al., 2025b). If the algebraic multiplicity of  $\lambda$  is greater than its geometric multiplicity, the definition of modal functional observability involves the Jordan normal form of  $A$  and the associated observability matrices; see Zhang et al. (2025b) for details. It has been shown in Zhang et al. (2025b) that  $(A, C, F)$  is functionally observable if and only if each eigenvalue of  $A$  is modal functionally observable. Below, we show that the output-reachability of  $X_F$  in  $\mathcal{G}(\bar{A}, \bar{C})$  implies that every nonzero eigenvalue of  $A$  is generically modal functionally observable, generalizing Hosoe and Matsumoto (1979, Theo 2).



**Proposition 3.** *If every  $x \in X_F$  is output-reachable, then for almost all realizations  $(A, C, F)$  of  $(\bar{A}, \bar{C}, \bar{F})$ , every nonzero eigenvalue of  $A$  is modal functionally observable, that is, there is a proper variety  $\mathbb{V} \subseteq \mathbb{R}^{n_A} \times \mathbb{R}^{n_C} \times \mathbb{R}^{n_F} \setminus \mathbb{V}$ , such that for any  $(A, C, F) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_C} \times \mathbb{R}^{n_F} \setminus \mathbb{V}$ ,  $\text{rank}[A - \lambda I; C; F] = \text{rank}[A - \lambda I; C]$ , where  $\lambda$  is any nonzero eigenvalue of  $A$ .*

**Proof.** Without loss of generality, assume that vertices  $X_1 \doteq \{x_1, \dots, x_{n_1}\}$  are output-reachable while  $X_2 \doteq X \setminus X_1 = \{x_{n_1+1}, \dots, x_n\}$  are not, where  $n_1 = |X_1|$ . If every  $x \in X_F$  is output-reachable, i.e.,  $X_F \subseteq X_1$ , then for all realizations  $(A, C, F)$  of  $(\bar{A}, \bar{C}, \bar{F})$ ,  $[A - \lambda I; C; F]$  has the following form

$$\begin{bmatrix} A_{11} - \lambda I & 0 \\ A_{21} & A_{22} - \lambda I \\ C_1 & 0 \\ F_1 & 0 \end{bmatrix},$$

where  $A_{11} \in \mathbb{R}^{n_1 \times n_1}$ ,  $A_{22} \in \mathbb{R}^{(n-n_1) \times (n-n_1)}$ , and  $[C_1, 0]$  and  $[F_1, 0]$  are partitioned in accordance with  $A$ . From Lemma 3, there is a proper variety  $\mathbb{V} \subseteq \mathbb{R}^{n_A} \times \mathbb{R}^{n_C} \times \mathbb{R}^{n_F}$  such that for every  $(A, C, F) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_C} \times \mathbb{R}^{n_F} \setminus \mathbb{V}$ , all nonzero eigenvalues of  $A$  (if any) are simple, and  $\text{rank}[A_{11} - \lambda I; C_1] = n_1$  for each nonzero eigenvalue  $\lambda$  of  $A_{11}$ . This leads to that  $\text{rank}[A_{11} - \lambda I; C_1] = n_1, \forall \lambda \neq 0$ . As a result, for all  $\lambda \in \mathbb{C} \setminus \{0\}$ ,

$$\text{rank} \begin{bmatrix} A_{11} - \lambda I & 0 \\ A_{12} & A_{22} - \lambda I \\ C_1 & 0 \\ F_1 & 0 \end{bmatrix} = \text{rank} \begin{bmatrix} A_{11} - \lambda I & 0 \\ A_{12} & A_{22} - \lambda I \\ C_1 & 0 \\ 0 & 0 \end{bmatrix},$$

which is due to the fact that  $\mathcal{R}(F_1) \subseteq \mathcal{R}([A_{11} - \lambda I; C_1])$ , with  $\mathcal{R}(M)$  denoting the space spanned by the rows of a matrix  $M$ . The above relation is exactly the definition of modal functional observability for a simple eigenvalue  $\lambda$  of  $A$ , leading to the proposed statement.  $\square$

**Theorem 2.** *Given a triple  $(\bar{A}, \bar{C}, \bar{F})$ , suppose  $\bar{A}$  is generically diagonalizable. Then, the following statements are equivalent:*

- (a)  $(\bar{A}, \bar{C}, \bar{F})$  is SFO.
- (b) (b1) Every  $x \in X_F$  is output-reachable, and (b2)  $\text{grank}[\bar{A}; \bar{C}] = \text{grank}[\bar{A}; \bar{C}; \bar{F}]$ .
- (c) (c1) Every  $x \in X_F$  is output-reachable, and (c2)  $\text{grank}[\bar{A}; \bar{C}] = \text{grank}[\bar{A}; \bar{C}; \bar{e}_i]$  for each  $x_i \in X_F$ .
- (d) (d1) Every  $x \in X_F$  is output-reachable, and (d2) no  $x \in X_F$  is contained in a minimal dilation of  $\mathcal{G}(\bar{A}, \bar{C})$ .

**Proof** (b) $\Rightarrow$ (a): From Proposition 3, condition (b1) implies that there is a proper variety  $\mathbb{V}_1$  such that for all  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}_1$ ,  $\text{rank}[A - \lambda I; C; F] = \text{rank}[A - \lambda I; C], \forall \lambda \in \mathbb{C} \setminus \{0\}$ . Condition (b2) implies that there is a proper variety  $\mathbb{V}_2$  such that for all  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}_2$ ,  $\text{rank}[A; C] = \text{rank}[A; C; F]$ . Since  $\bar{A}$  is generically diagonalizable, by definition, there exists a proper variety  $\mathbb{V}_3$  such that for all  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}_3$ ,  $A$  is diagonalizable. Those conditions together yield that for the proper variety  $\mathbb{V}_4 = \mathbb{V}_1 \cup \mathbb{V}_2 \cup \mathbb{V}_3$ , all realizations  $(A, C, F) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_C} \times \mathbb{R}^{n_F} \setminus \mathbb{V}_4$  satisfy that  $A$  is diagonalizable, and

$$\text{rank}[A - \lambda I; C; F] = \text{rank}[A - \lambda I; C], \forall \lambda \in \mathbb{C},$$

which indicates that  $(A, C, F)$  is functionally observable from Lemma 1. By definition,  $(\bar{A}, \bar{C}, \bar{F})$  is SFO.

(a) $\Rightarrow$ (b): If (b1) is not satisfied, without loss of generality, assume that  $X_1 = \{x_1, \dots, x_{n_1}\}$  is the set of output-reachable vertices, and  $X_2 = X \setminus X_1 = \{x_{n_1+1}, \dots, x_n\}$ . It follows that  $X_F \cap X_2 \neq \emptyset$ . Accordingly, partition  $(\bar{A}, \bar{C}, \bar{F})$  as

$$\begin{bmatrix} \bar{A} \\ \bar{C} \\ \bar{F} \end{bmatrix} = \begin{bmatrix} \bar{A}_{11} & 0 \\ \bar{A}_{12} & \bar{A}_{22} \\ \bar{C}_1 & 0 \\ \bar{F}_1 & \bar{F}_2 \end{bmatrix},$$

with  $\bar{A}_{11} \in \{0, *\}^{n_1 \times n_1}$ ,  $\bar{A}_{22} \in \{0, *\}^{(n-n_1) \times (n-n_1)}$ , and  $\bar{F}_2 \neq 0$ . Then,  $\text{grank } O(\bar{A}, \bar{C}) = \text{grank } O(\bar{A}_{11}, \bar{C}_1)$ ,  $\text{grank } O(\bar{A}, [\bar{C}; \bar{F}]) \geq \text{grank } O(\bar{A}_{11}, \bar{C}_1) + \text{grank } O(\bar{A}_{22}, \bar{F}_2)$ . As  $\bar{F}_2 \neq 0$ ,  $\text{grank } O(\bar{A}_{22}, \bar{F}_2) \geq 1$ , yielding  $\text{grank } O(\bar{A}, [\bar{C}; \bar{F}]) > \text{grank } O(\bar{A}, \bar{C})$ . By Lemma 2,  $(\bar{A}, \bar{C}, \bar{F})$  is not SFO, which demonstrates the necessity of (b1) for (a). The necessity of (b2) for (a) is obvious since condition (3) should hold when  $\lambda = 0$ .

To show (b) $\Leftrightarrow$ (c), it suffices to show (b2) $\Leftrightarrow$ (c2). The direction (b2) $\Rightarrow$ (c2) is obvious by noting that  $\forall x_i \in X_F$ ,  $\text{grank}[\bar{A}; \bar{C}; \bar{F}] \geq \text{grank}[\bar{A}; \bar{C}; \bar{e}_i] \geq \text{grank}[\bar{A}; \bar{C}]$ .

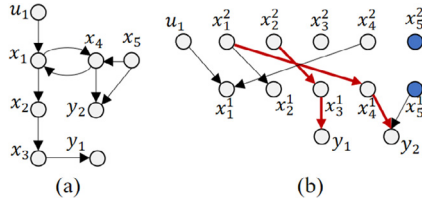
(c2) $\Rightarrow$ (b2): By the property of rank function,<sup>3</sup> if  $\text{grank}[\bar{A}; \bar{C}] = \text{grank}[\bar{A}; \bar{C}; \bar{e}_i] = \text{grank}[\bar{A}; \bar{C}; \bar{e}_j]$  whenever  $i \neq j$ , then  $\text{grank}[\bar{A}; \bar{C}] = \text{grank}[\bar{A}; \bar{C}; \bar{e}_i; \bar{e}_j]$ . Repeat this procedure, and we get  $\text{grank}[\bar{A}; \bar{C}] = \text{grank}[\bar{A}; \bar{C}; \bar{I}_{X_F}]$ . Let  $X_F^c = X \setminus X_F$ . Observe that  $\text{grank}[\bar{A}; \bar{C}; \bar{I}_{X_F}] = \text{grank}[\bar{A}; \bar{C}](\cdot, X_F^c) + |X_F|$ , and  $\text{grank}[\bar{A}; \bar{C}] \leq \text{grank}[\bar{A}; \bar{C}; \bar{F}] \leq \text{grank}[\bar{A}; \bar{C}](\cdot, X_F^c) + |X_F|$ . It follows that  $\text{grank}[\bar{A}; \bar{C}] = \text{grank}[\bar{A}; \bar{C}; \bar{F}]$ , yielding (c2).

To show (c) $\Leftrightarrow$ (d), it suffices to show (c2) $\Leftrightarrow$ (d2), which has been proved in Montanari et al. (2022, Lems 2, 5).  $\square$

**Remark 1.** It is worth mentioning that condition (d) was posited as both necessary and sufficient for the SFO of a general structured system in Montanari et al. (2022). However, it appears there might have been some misunderstanding in treating the PBH-like necessary condition (condition (3)) as a sufficient condition for functional observability in the general case. This perspective, which originally appeared in Moreno (2001) and Jennings et al. (2011), was not clarified until recently (Darouach & Fernando, 2025; Zhang et al., 2025b). Theorem 2 suggests that the conditions obtained in Montanari et al. (2022) are applicable to the class of generically diagonalizable systems. It is essential to underscore several key points:

- Firstly, Montanari et al. (2022) derived condition (d) under a different definition of SFO compared to ours. In their definition, a structured system is considered SFO if there exists a realization that is functionally observable, rather than in the generic case. As outlined in Definition 3 and demonstrated in Example 1, these two definitions are not equivalent, even for generically diagonalizable systems (notice that  $\bar{A}$  in Example 1 is generically diagonalizable). This variance in the definitions of SFO leads to distinct derivations of SFO criteria.
- Secondly, condition (d) given in Montanari et al. (2022) was predicated on an additional assumption – that each row of  $F$  contains only one nonzero entry. This assumption is not needed here.
- Thirdly, there is no guarantee that the graph-theoretic condition (d) is equivalent to condition (3) in the generic sense if no constraint is imposed on  $\bar{A}$  (see Remark 7 of Zhang et al. (2025b)). Building on Proposition 3, it may be possible to broaden Theorem 2 to encompass more general systems. However, the outcomes would be contingent on  $\bar{A}, \bar{C}$ , and  $\bar{F}$ , rather than solely on  $\bar{A}$ . In particular, Montanari et al. (2025, Theo 1) recently proposed additional conditions to ensure the equivalence between condition (3) of Lemma 1

<sup>3</sup> The generic rank function has the submodularity property (Murota, 2009), that is, given  $\bar{M} \in \{0, *\}^{m \times n}$ , for any  $S_1, S_2 \subseteq [m]$ ,  $\text{grank } \bar{M}(S_1, \cdot) + \text{grank } \bar{M}(S_2, \cdot) \geq \text{grank } \bar{M}(S_1 \cup S_2, \cdot) + \text{grank } \bar{M}(S_1 \cap S_2, \cdot)$ .



**Fig. 2.**  $\mathcal{G}(\bar{A}, \bar{B}, \bar{C})$  and  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  in Example 2. Vertices in blue correspond to input-unreachable vertices in  $\mathcal{G}(\bar{A}, \bar{B})$ . Bold red edges constitute a linking of size 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and condition (ii) of Lemma 2 for a numerical system, depending on the matrices  $A$  and  $F$  without requiring the diagonalizability of  $A$ . These results may be promising in deriving graph-theoretic conditions for SFO like Theorem 2.

Compared to Lemma 2 or the graph-theoretic conditions in Zhang et al. (2025b), Theorem 2 stands out for its remarkable simplicity as it does need the global information of  $(\bar{A}, \bar{C}, \bar{F})$  (such as the generic dimension of unobservable subspaces, i.e.,  $n - \text{grank } O(\bar{A}, \bar{C})$ ). This notable feature enables the derivation of a closed-form solution for the minimal sensor placement problem within the class of generically diagonalizable systems, a problem that remains open otherwise (see Section 7).

## 6. Polynomial-time verifiable criteria for SOC

In this section, we identify a class of systems for which the SOC can be checked in polynomial time and highlight that generically diagonalizable systems fall into this class.

Let  $X_r$  be the set of input-reachable state vertices in  $\mathcal{G}(\bar{A}, \bar{B})$ . Let  $\bar{A}_r \in \{0, *\}^{n \times n}$  be obtained from  $\bar{A}$  by replacing its rows and columns corresponding to  $X \setminus X_r$  with zero blocks. Construct a directed graph  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  associated with  $(\bar{A}_r, \bar{B}, \bar{C})$  as follows: the vertex set is  $X^2 \cup U^2 \cup X^1 \cup Y^0$ , with  $X^2 = \{x_1^2, \dots, x_n^2\}$ ,  $U^2 = \{u_1, \dots, u_m\}$ ,  $X^1 = \{x_1^1, \dots, x_n^1\}$ , and  $Y^0 = \{y_1, \dots, y_p\}$ , and the edge set is  $E_{21} \cup E_{10}$ , with  $E_{21} = \{(x_i^2, x_j^1) : [\bar{A}_r]_{ji} \neq 0\} \cup \{(u_i, x_j^1) : [\bar{B}]_{ji} \neq 0\}$  and  $E_{10} = \{(x_i^1, y_j) : [\bar{C}]_{ji} \neq 0\}$ . A linking of size  $l$  from  $X^2 \cup U^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  is a collection of  $l$  vertex-disjoint simple paths, in which each path has its initial vertex in  $X^2 \cup U^2$  and terminal vertex in  $Y^0$ . Below, to illustrate these definitions, we provide an example.

**Example 2.** Consider a system  $(\bar{A}, \bar{B}, \bar{C})$  whose digraph  $\mathcal{G}(\bar{A}, \bar{B}, \bar{C})$  is given in Fig. 2(a). From it, we known state vertices  $X_r = \{x_1, x_2, x_3, x_4\}$  are input-reachable, while  $x_5$  is not. Accordingly, the corresponding  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  is given in Fig. 2(b), where the edge  $(x_5^2, x_4^1)$  does not exist since  $x_5$  is not input-reachable. Two disjoint paths  $(x_1^2, x_4^1, y_2)$  and  $(x_2^2, x_3^1, y_1)$  constitute a linking of size 2 from  $X^2 \cup U^2$  to  $Y^0$ .

**Theorem 3.** Given that  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank } Q(\bar{A}, \bar{B})$ ,  $(\bar{A}, \bar{B}, \bar{C})$  is SOC if and only if there is a linking of size  $p$  from  $X^2 \cup U^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$ .

**Proof.** From Papadimitriou and Tsitsiklis (1984, Theo. 6), there is a linking of size  $p$  from  $X^2 \cup U^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  if and only if  $\text{grank } \bar{C}[\bar{A}_r, \bar{B}] = p$ . To proceed with the proof, first consider the case where all state vertices are input-reachable, i.e., the case  $\bar{A}_r = \bar{A}$ . As  $\text{grank}[\bar{A}, \bar{B}] = \text{grank } Q(\bar{A}, \bar{B})$ , there is a proper variety  $\mathbb{V} \subseteq \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C}$  such that for any  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}$ , it holds that  $\text{rank}[A, B] = \text{rank } Q(A, B)$ .

Observe that for any  $z \in \mathbb{C}^{1 \times n}$  satisfying  $z[A, B] = 0$ , we have  $zA^k B = 0$  for  $k = 0, 1, \dots, n-1$ , which yields  $zQ(A, B) = 0$ . That is, upon letting  $\mathcal{N}(\cdot)$  be the left-null space of a matrix, we have  $\mathcal{N}([A, B]) \subseteq \mathcal{N}(Q(A, B))$ . Since  $\text{rank}[A, B] = \text{rank } Q(A, B)$ , which implies  $\dim \mathcal{N}([A, B]) = \dim \mathcal{N}(Q(A, B))$  ( $\dim(\cdot)$  takes the dimension of a subspace), we arrive at

$$\mathcal{N}([A, B]) = \mathcal{N}(Q(A, B)). \quad (4)$$

If  $(A, B, C)$  is not output controllable, there exists some nonzero  $z \in \mathbb{C}^{1 \times p}$  such that  $zCQ(A, B) = 0$ . That is,  $zC \in \mathcal{N}(Q(A, B))$ . Relation (4) yields  $zC[A, B] = 0$ , which implies  $\text{rank } C[A, B] < p$ . Conversely, if  $C[A, B]$  does not have full row rank, there exists some nonzero  $z' \in \mathbb{C}^{1 \times p}$  such that  $z'C[A, B] = 0$ . This means  $z'C \in \mathcal{N}([A, B])$ , leading to  $z'CQ(A, B) = 0$  from (4), i.e.,  $(A, B, C)$  is not output controllable. The above analysis shows that for any  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}$ ,  $(A, B, C)$  is output controllable if and only if  $\text{rank } C[A, B] = p$ . As a result, if  $\text{grank } \bar{C}[\bar{A}, \bar{B}] = p$ , there is a proper variety  $\mathbb{V}' \subseteq \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C}$  such that for any  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}'$ ,  $\text{rank } C[A, B] = p$ . Moreover, for each  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus (\mathbb{V} \cup \mathbb{V}')$ ,  $(A, B, C)$  is output controllable. Conversely, if  $\text{grank } \bar{C}[\bar{A}, \bar{B}] < p$ , then for all  $(A, B, C) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B} \times \mathbb{R}^{n_C} \setminus \mathbb{V}$ ,  $(A, B, C)$  is not output controllable since  $\text{rank } C[A, B] < p$ . By definition,  $(\bar{A}, \bar{B}, \bar{C})$  is SOC if and only if  $\text{grank } \bar{C}[\bar{A}, \bar{B}] = p$ .

Now consider the case where not all state vertices are input-reachable in  $\mathcal{G}(\bar{A}, \bar{B})$ . Assume  $X_r \subseteq X$  is the set of input-reachable vertices in  $\mathcal{G}(\bar{A}, \bar{B})$ . Without losing generality, suppose  $[\bar{A}, \bar{B}]$  and  $A_r$  have the following forms, respectively (if not, one can re-order the state vertices):

$$[\bar{A}, \bar{B}] = \begin{bmatrix} \bar{A}_{11} & \bar{A}_{12} & \bar{B}_1 \\ 0 & \bar{A}_{22} & 0 \end{bmatrix}, A_r = \begin{bmatrix} \bar{A}_{11} & 0 \\ 0 & 0 \end{bmatrix}, \quad (5)$$

where  $\bar{A}_{11} \in \{0, *\}^{|X_r| \times |X_r|}$ ,  $\bar{A}_{22} \in \{0, *\}^{|X \setminus X_r| \times |X \setminus X_r|}$ , and  $\bar{B}_1 \in \{0, *\}^{|X_r| \times m}$ . Moreover, partition  $\bar{C}$  as  $\bar{C} = [\bar{C}_1, \bar{C}_2]$  with  $\bar{C}_1 \in \{0, *\}^{p \times |X_r|}$  and  $\bar{C}_2 \in \{0, *\}^{p \times |X \setminus X_r|}$ . It turns out that

$$Q(\bar{A}, \bar{B}) = \begin{bmatrix} \bar{B}_1 & \bar{A}_{11}\bar{B}_1 & \dots & \bar{A}_{11}^{n-1}\bar{B}_1 \\ 0 & 0 & \dots & 0 \end{bmatrix}, \bar{A}_r = \begin{bmatrix} \bar{A}_{11} & 0 \\ 0 & 0 \end{bmatrix}. \quad (6)$$

Therefore,  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank } Q(\bar{A}, \bar{B})$  if and only if  $\text{grank}[\bar{A}_{11}, \bar{B}_1] = \text{grank } Q(\bar{A}_{11}, \bar{B}_1)$ . Moreover,  $\text{grank } \bar{C}Q(\bar{A}, \bar{B}) = \text{grank } \bar{C}_1 Q(\bar{A}_{11}, \bar{B}_1)$  and  $\text{grank } \bar{C}[\bar{A}_r, \bar{B}] = \text{grank } \bar{C}_1[\bar{A}_{11}, \bar{B}_1]$ . Observe that every state vertex of  $(\bar{A}_{11}, \bar{B}_1)$  is input-reachable. By adopting the above SOC condition on  $(\bar{A}_{11}, \bar{B}_1, \bar{C}_1)$ , we obtain the required result.  $\square$

In light of Theorem 3, whenever  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank } Q(\bar{A}, \bar{B})$ , we can check the SOC of  $(\bar{A}, \bar{B}, \bar{C})$  by verifying whether a linking of size  $p$  exists in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$ . All these conditions can be verified in polynomial time. To be specific,  $\text{grank}[\bar{A}_r, \bar{B}]$  is obtainable by computing the maximum size of matchings in the bipartite graph associated with  $[\bar{A}_r, \bar{B}]$ . And  $\text{grank } Q(\bar{A}, \bar{B})$  equals the maximum number of state vertices that can be covered by an input cactus configuration in  $\mathcal{G}(\bar{A}, \bar{B})$  (Hosoe, 1980), which can be determined via weighted maximum matching algorithms (Murota & Poljak, 1990, Theo. 6). Moreover, the maximum size of linkings in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  can be obtained via the maximum flow algorithm in  $O(n^{2.5})$  time (Papadimitriou and Tsitsiklis 1984, Theo. 6; see also Section 7.2).

**Example 3 (Example 2 Continued).** Consider the system  $(\bar{A}, \bar{B}, \bar{C})$  given in Example 2. Since the maximum number of input-reachable state vertices in a collection of disjoint input stems and cycles is 3, we have  $\text{grank } Q(\bar{A}, \bar{B}) = 3$ . Moreover,  $\text{grank}[\bar{A}_r, \bar{B}] = 3$ , leading to  $\text{grank } Q(\bar{A}, \bar{B}) = \text{grank}[\bar{A}_r, \bar{B}]$ . As there is a linking of size 2 from  $X^2 \cup U^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  (highlighted in bold red lines in Example 2(b)), Theorem 3 yields that  $(\bar{A}, \bar{B}, \bar{C})$  is SOC.



We emphasize that  $\bar{A}$  in [Example 3](#) is not generically diagonalizable and the condition  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank} Q(\bar{A}, \bar{B})$  depend on both  $\bar{A}$  and  $\bar{B}$ . In what follows, we show that for the class of generically diagonalizable systems, this condition always holds irrespective of  $\bar{B}$ .

**Corollary 4.** *If  $\bar{A}$  is generically diagonalizable, then  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank} Q(\bar{A}, \bar{B})$  for any  $\bar{B} \in \{0, *\}^{n \times m}$ . Consequently,  $(\bar{A}, \bar{B}, \bar{C})$  is SOC if and only if there is a linking of size  $p$  from  $X^2 \cup U^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$ .*

**Proof.** We first consider the case where every state vertex in  $\mathcal{G}(\bar{A}, \bar{B})$  is input-reachable, i.e., the case  $\bar{A}_r = \bar{A}$ . By [Lemma 3](#), for a generically diagonalizable  $\bar{A}$ , there is a proper variety  $\mathbb{V}$  such that for each  $(A, B) \in \mathbb{R}^{n \times n} \times \mathbb{R}^{n \times m} \setminus \mathbb{V}$ , it holds that (i)  $A$  is diagonalizable and (ii) each nonzero eigenvalue of  $A$  is simple and controllable for system  $(A, B)$ . We are to show that  $\text{rank}[A, B] = \text{rank} Q(A, B)$ . Suppose  $A$  has  $q$  distinct eigenvalues  $\lambda_1, \dots, \lambda_q$ , in which  $\lambda_1 = 0$ ,  $\lambda_2, \dots, \lambda_q \neq 0$ . Moreover, let  $z_i \in \mathbb{C}^{n_i \times n}$  consist of a set of linearly independent row vectors spanning the left eigenspace associated with  $\lambda_i$ , where  $n_i$  is the geometric multiplicity of  $\lambda_i$ , and  $Z$  be stacked by  $z_1, \dots, z_q$  from up to down (if all eigenvalues of  $A$  are nonzero, then  $\lambda_1 = 0$  does not exist and  $n_1 = 0$ , under which all the subsequent statements remain valid). Then,  $ZAZ^{-1} = \text{diag}\{\lambda_i I_{n_i} |_{i=1}^q\}$ . According to [Olshevsky \(2014, Lem. 8\)](#), we have

$$\begin{aligned} \text{rank} Q(A, B) &= \sum_{i=1}^q \text{rank} Q(\lambda_i I_{n_i}, z_i B) \\ &= \text{rank } z_1 B + q - 1, \end{aligned}$$

which comes from the fact that eigenvalues  $\lambda_2, \dots, \lambda_q$  are controllable, leading to  $z_i B \neq 0$  from the PBH test for  $i = 2, \dots, q$ . Observe that

$$\begin{aligned} \text{rank}[A, B] &= \text{rank}[ZAZ^{-1}, ZB] \\ &= \text{rank}[\text{diag}\{\lambda_i I_{n_i} |_{i=1}^q\}, \text{col}\{z_i |_{i=1}^q\} B] = q - 1 + \text{rank } z_1 B. \end{aligned}$$

This leads to  $\text{rank} Q(A, B) = \text{rank}[A, B]$ . Since this relation holds for almost all realizations  $(A, B)$  of  $(\bar{A}, \bar{B})$ , we reach  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank} Q(\bar{A}, \bar{B})$ .

If not every state vertex is input-reachable in  $\mathcal{G}(\bar{A}, \bar{B})$ , without losing generality, assume that  $[\bar{A}, \bar{B}]$  has the form as [\(5\)](#). From [Proposition 1](#), since  $\bar{A}$  is generically diagonalizable, so is  $\bar{A}_{11}$ . Using the above result on  $(\bar{A}_{11}, \bar{B}_1)$ , we have  $\text{grank}[\bar{A}_r, \bar{B}] = \text{grank}[\bar{A}_{11}, \bar{B}_1] = \text{grank} Q(\bar{A}_{11}, \bar{B}_1) = \text{grank} Q(\bar{A}, \bar{B})$ , where the last equality follows the same line as [\(6\)](#). The remaining statement is immediate from [Theorem 3](#).  $\square$

**Remark 2.** By inspection, there is a linking of size  $p$  from  $X^2 \cup U^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}_r, \bar{B}, \bar{C})$  if and only if there is a subset  $X_S \subseteq X$  with  $|X_S| = p$  such that  $\text{grank}[\bar{A}_r, \bar{B}](X_S, \cdot) = p$  and  $\text{grank} \bar{C}(\cdot, X_S) = p$ . While a similar condition is given in [Li et al. \(2021, Theo 2\)](#) for the SOC of undirected networks with the symmetric weight constraint  $A = A^T$ , our criteria are applied to different system classes. Notably, based on [Corollary 3](#), by comparing [Li et al. \(2021, Theo 2\)](#) and [Corollary 4](#), it turns out that an undirected network with the symmetric weight constraint is SOC if and only if it is SOC without such a constraint. This is consistent with [Zhang et al. \(2025a\)](#), where this result is derived using a purely graph-theoretic approach. The dilation-based structural target controllability condition presented in [Montanari et al. \(2023, Theo 2\)](#) is equivalent to [Corollary 4](#) when each row of  $\bar{C}$  has at most one nonzero entry. However, they have not defined generic diagonalizability, and their utilization of the

diagonalizability condition is implicit. Besides, they require that each column of  $\bar{B}$  contains at most one nonzero entry, which is unnecessary here.

**Remark 3 (Duality Between SFO and SOC).** Although there is no direct duality between SFO and SOC ([Iudice et al., 2019](#)), i.e., the SFO of  $(\bar{A}, \bar{C}, \bar{F})$  is generally not equivalent to the SOC of  $(\bar{A}^T, \bar{C}^T, \bar{F})$ , [Montanari et al. \(2023\)](#) introduced some weak and strong dualities between them. Here, “weak” duality means that one property indicates the other, but the reverse may not hold, while “strong” duality means that one property implies the other and vice versa under certain conditions. For example, the SFO of  $(\bar{A}, \bar{C}, \bar{F})$  implies the SOC of  $(\bar{A}^T, \bar{C}^T, \bar{F})$  ([Montanari et al., 2023, Theo 4](#)), although the inverse may not hold. For more details, see [Montanari et al. \(2023, 2024, 2025\)](#).

## 7. The minimal sensor and actuator placement problems

### 7.1. Minimal sensor placement for SFO

The minimal sensor placement problem for SFO can be formulated as

$$\begin{aligned} \min_{\bar{C} \in \{0, *\}^{p \times n}} \quad & p \\ \text{s.t.} \quad & (\bar{A}, \bar{C}, \bar{F}) \text{ SFO.} \end{aligned} \quad (\mathcal{P}_1)$$

Problem  $\mathcal{P}_1$  ( $\mathcal{P}_1$  in short) asks to minimize the number of linear functions of states needed to be measured to estimate  $z(t)$  ( $z(t)$  is a linear combination of state variables with generic coefficients). Notice that there is no constraint on the structure of the available  $\bar{C}$ . It has been shown in [Zhang et al. \(2025b, Prop 5\)](#) that if prior structure constraint is imposed on  $\bar{C}$ , such as the dedicated output constraint (i.e., each sensor measures only one state variable),  $\mathcal{P}_1$  is NP-hard even for systems with self-loops in all states. This means even within the class of generically diagonalizable systems, imposing prior structure constraint to  $\bar{C}$  will lead to the NP-hardness of  $\mathcal{P}_1$ . Nevertheless, the unconstrained case remains open.

**Remark 4 (Non-monotonicity of SFO).** It is noteworthy that the challenge of  $\mathcal{P}_1$  lies in the ‘non-monotonicity’ of SFO w.r.t. edge additions, that is, adding additional edges from state vertices to the existing sensors may destroy SFO. To illustrate this, consider two triples  $(\bar{A}, \bar{C}_i, \bar{F})$ ,  $i \in \{1, 2\}$ , in which  $\bar{A} = 0_{2 \times 2}$ ,  $\bar{C}_1 = [*, 0]$ ,  $\bar{C}_2 = [*, *]$ , and  $\bar{F} = [*, 0]$ .  $\mathcal{G}(\bar{A}, \bar{C}_2)$  is obtained from  $\mathcal{G}(\bar{A}, \bar{C}_1)$  by adding an edge from the second state vertex to the unique sensor. It can be verified that  $(\bar{A}, \bar{C}_1, \bar{F})$  is SFO while  $(\bar{A}, \bar{C}_2, \bar{F})$  not.

[Theorem 2](#) implies that the SFO of  $(\bar{A}, \bar{C}, \bar{F})$  for a generically diagonalizable system depends only on  $X_F$ , i.e., the functional state set (irrespective of how they are summed in  $z(t)$ ). To derive solutions for the minimal sensor problem, we first generalize this result to general systems, shown as follows.

**Corollary 5.** *Given a triple  $(\bar{A}, \bar{C}, \bar{F})$ , the following statements are equivalent:*

- (a)  $(\bar{A}, \bar{C}, \bar{F})$  is SFO.
- (b)  $(\bar{A}, \bar{C}, \bar{I}_{X_F})$  is SFO.
- (c)  $(\bar{A}, \bar{C}, \bar{e}_i)$  is SFO for each  $x_i \in X_F$ .
- (d)  $(\bar{A}, \bar{C}, \bar{I}_{X_S})$  is SFO for every non-empty  $X_S \subseteq X_F$ .

**Proof.** By regarding  $O(\bar{A}, \bar{C})$  as  $[\bar{A}; \bar{C}]$  in the proof of ‘(b)  $\Leftrightarrow$  (c)’ of [Theorem 2](#), we can establish that  $\text{grank}[O(\bar{A}, \bar{C}); \bar{F}] = \text{grank}[O(\bar{A}, \bar{C})]$  if and only if  $\text{grank}[O(\bar{A}, \bar{C})] = \text{grank}[O(\bar{A}, \bar{C}); \bar{e}_i]$  for each  $x_i \in X_F$ . The required result then follows from (ii) of [Lemma 2](#).  $\square$

### 7.1.1. Generically diagonalizable system case

We give a closed-form solution as well as a weighted maximum matching based algorithm for  $\mathcal{P}_1$  in the class of generically diagonalizable systems.

**Theorem 4.** Given a pair  $(\bar{A}, \bar{F})$ , suppose  $\bar{A}$  is generically diagonalizable. Then, the optimal value  $p^*$  to  $\mathcal{P}_1$  is

$$p^* = \max \{ \text{grank} [\bar{A}; \bar{I}_{X_F}] - \text{grank} \bar{A}, 1 \}. \quad (7)$$

Moreover, let  $X_F^u$  be a subset of  $X_F$  with the maximum cardinality that is right-unmatched in a maximum matching of  $\mathcal{B}(\bar{A})$ . Then,  $p^* = \max\{|X_F^u|, 1\}$ , and Algorithm 1 can find an optimal solution to  $\mathcal{P}_1$ .

**Proof.** According to Corollary 5,  $p^*$  is the optimal value to  $\mathcal{P}_1$  on  $(\bar{A}, \bar{F})$  if and only if  $p^*$  is the optimal value of  $\mathcal{P}_1$  on  $(\bar{A}, \bar{I}_{X_F})$ . Suppose  $\bar{C}_f$  is a feasible solution to  $\mathcal{P}_1$ . Theorem 2 yields  $\text{grank} [\bar{A}; \bar{C}_f; \bar{I}_{X_F}] = \text{grank} [\bar{A}; \bar{C}_f]$ . This implies  $\text{grank} [\bar{A}; \bar{C}_f] \geq \text{grank} [\bar{A}; \bar{I}_{X_F}]$ . Since  $\text{grank} [\bar{A}; \bar{C}_f] \leq \text{row}(\bar{C}_f) + \text{grank} \bar{A}$ , we have  $\text{row}(\bar{C}_f) \geq \text{grank} [\bar{A}; \bar{I}_{X_F}] - \text{grank} \bar{A}$ , which indicates (7) is a lower bound of  $\mathcal{P}_1$ .

We are to show that the lower bound (7) can be achieved via Algorithm 1. By the construction of the weighted bipartite graph  $\mathcal{B}(\bar{A})$ , it is obvious that  $X_F^u$  is a subset of  $X_F$  with the maximum cardinality that is right-unmatched in a maximum matching of  $\mathcal{B}(\bar{A})$ . We now show that  $|X_F^u| = \text{grank} [\bar{A}; \bar{I}_{X_F}] - \text{grank} \bar{A}$ . To show this, from the property of rank function (cf. Murota 2009, Prop. 2.1.3), we know that there exist  $S_1 \subseteq [n]$  and  $X_2 \subseteq X_F$  satisfying  $\text{grank} \bar{A}(S_1, X \setminus X_2) = |S_1| = \text{grank} \bar{A}$ ,  $\text{grank} \bar{I}_{X_F}(:, X_2) = |X_2|$  such that  $\text{grank} [\bar{A}; \bar{I}_{X_F}] = \text{grank} \bar{A}(S_1, :) + \text{grank} \bar{I}_{X_F}(:, X_2) = \text{grank} \bar{A}(S_1, X \setminus X_2) + \text{grank} \bar{I}_{X_F}(:, X_2) = |S_1| + |X_2|$ . As a result,  $\text{grank} [\bar{A}; \bar{I}_{X_F}] - \text{grank} \bar{A} = |X_2|$  and  $X_2 \subseteq X_F$  is right-unmatched in some maximum matching of  $\mathcal{B}(\bar{A})$  (noting that  $\bar{A}(S_1, X \setminus X_2)$  corresponds to a maximum matching of  $\mathcal{B}(\bar{A})$ ). Suppose  $X_2$  is not of the maximum cardinality over all such possible sets, that is, there is a maximum matching  $\mathcal{M}'$  of  $\mathcal{B}(\bar{A})$  associated with which the set of right-unmatched vertices belonging to  $X_F$ , denoted by  $X_2'$ , has a larger size than  $|X_2|$ . Then,  $\text{grank} [\bar{A}; \bar{I}_{X_F}] \geq |S_1| + |X_2'| > |S_1| + |X_2|$ , since adding  $|X_2'|$  edges to  $\mathcal{M}'$  will form a new matching in  $\mathcal{B}(\bar{A}; \bar{I}_{X_F})$  with size  $|S_1| + |X_2'|$ , contradicting the fact that  $\text{grank} [\bar{A}; \bar{I}_{X_F}] = |S_1| + |X_2|$ .

Let  $\bar{C}'$  denote the matrix obtained after Step 3, and  $\bar{C}$  the matrix obtained via Algorithm 1. Consider the case  $\text{grank} [\bar{A}; \bar{I}_{X_F}] - \text{grank} \bar{A} \geq 1$ . From the above analysis, we have  $\text{grank} [\bar{A}; \bar{C}'] = \text{grank} [\bar{A}; \bar{I}_{X_F}] = \text{grank} \bar{A} + p^*$ . Notice that each row of  $\bar{C}'$  is also a row of  $\bar{I}_{X_F}$  with only one nonzero entry, leading to  $\text{grank} [\bar{A}; \bar{C}'; \bar{I}_{X_F}] = \text{grank} [\bar{A}; \bar{I}_{X_F}]$ . Hence,  $\text{grank} [\bar{A}; \bar{C}'] = \text{grank} [\bar{A}; \bar{C}'; \bar{I}_{X_F}]$ . Let  $X_F^c = X \setminus X_F$ . Observe that all nonzero columns of  $\bar{C}$  correspond to state vertices contained in  $X_F$ , yielding  $\text{grank} [\bar{A}; \bar{C}; \bar{I}_{X_F}] = |X_F| + \text{grank} \bar{A}_{X_F^c}$ , where  $\bar{A}_{X_F^c} \doteq \bar{A}(:, X_F^c)$ . On the other hand, by the construction, it holds that  $\text{grank} [\bar{A}; \bar{C}'] \leq \text{grank} [\bar{A}; \bar{C}] \leq |X_F| + \text{grank} \bar{A}_{X_F^c}$ . Since  $\text{grank} [\bar{A}; \bar{C}'] = \text{grank} [\bar{A}; \bar{I}_{X_F}] = |X_F| + \text{grank} \bar{A}_{X_F^c}$ , it follows that  $\text{grank} [\bar{A}; \bar{C}] = |X_F| + \text{grank} \bar{A}_{X_F^c}$ . We therefore reach  $\text{grank} [\bar{A}; \bar{C}; \bar{I}_{X_F}] = |X_F| + \text{grank} \bar{A}_{X_F^c} = \text{grank} [\bar{A}; \bar{C}]$ . As all  $x_i \in X_F$  are output-reachable in  $\mathcal{G}(\bar{A}, \bar{C})$ , Theorem 2 indicates that  $(\bar{A}, \bar{C}, \bar{F})$  is SFO. If  $\text{grank} [\bar{A}; \bar{I}_{X_F}] - \text{grank} \bar{A} = 0$ , since all nonzero columns of  $\bar{C}$  correspond to exactly  $X_F$ , we get  $\text{grank} [\bar{A}; \bar{C}; \bar{I}_{X_F}] = |X_F| + \text{grank} \bar{A}_{X_F^c} = \text{grank} [\bar{A}; \bar{I}_{X_F}]$ . As  $\text{grank} [\bar{A}; \bar{I}_{X_F}] = \text{grank} \bar{A}$ , it holds  $\text{grank} [\bar{A}; \bar{C}] = \text{grank} \bar{A} = \text{grank} [\bar{A}; \bar{C}; \bar{I}_{X_F}]$ . Since all  $x_i \in X_F$  are output-reachable,  $(\bar{A}, \bar{C}, \bar{F})$  is SFO.  $\square$

**Algorithm 1 :** A weighted maximum matching based algorithm for  $\mathcal{P}_1$  in generically diagonalizable systems

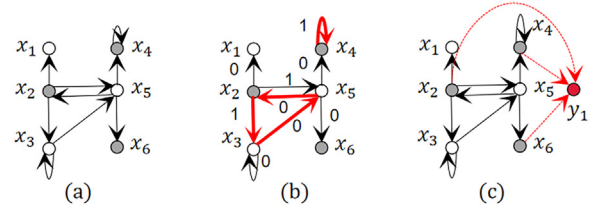
**Require:**  $\bar{A}$  and  $\bar{F}$  with  $\bar{A}$  generically diagonalizable.

**Ensure:** An optimal solution to  $\mathcal{P}_1$ .

- 1: Construct a weighted bipartite graph  $\mathcal{B}(\bar{A})$  by assigning edge cost  $w(e) : E_{XX} \rightarrow \{0, 1\}$  as

$$w(e) = \begin{cases} 1, & \text{if } e = (x_i, x_j), x_i \in X_F \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

- 2: Find a minimum weight maximum matching  $\mathcal{M}$  of  $\mathcal{B}(\bar{A})$ . Let  $X_{\mathcal{M}} = \{x_i : (x_i, x_j) \in \mathcal{M}\}$  be the set of right-matched vertices in  $\mathcal{M}$ ,  $X_S = X_F \cap X_{\mathcal{M}}$ , and  $X_F^u = X_F \setminus X_S$ .
- 3: Construct a  $\max\{1, |X_F^u|\} \times n$  matrix  $\bar{C}$  such that  $\bar{C}_{ki} = *$  if  $x_i$  is the  $k$ th element in  $X_F^u$ ,  $k = 1, \dots, |X_F^u|$ .
- 4: Per  $x_i \in X_S$ , choose one  $j \in \{1, \dots, \max\{1, |X_F^u|\}\}$  and let  $\bar{C}_{ji} = *$ .
- 5: Return  $\bar{C}$ .



**Fig. 3.** Example of the procedure of Algorithm 1 applied to the system digraph  $\mathcal{G}(\bar{A})$  in (a), where  $\bar{A}$  is generically diagonalizable. Nodes in gray represent functional states. In (b), the number on each edge is the cost assigned to it (see (8)), and bold red edges form a minimum weight maximum matching  $\mathcal{M}$  of  $\mathcal{B}(\bar{A})$ . Accordingly,  $X_S = \{x_2, x_4\}$  and  $X_F^u = \{x_6\}$ , resulting in  $p^* = 1$ . In (c), the dotted red edges represent the obtained solution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Remark 5.** From the proof of Theorem 4, it is easy to see that in Step 4 of Algorithm 1, instead of making  $\bar{C}_{ji} = *$  for each  $x_i \in X_S$ , it suffices to choose those  $x_i \in X_S$ , making  $\bar{C}_{ji} = *$ , such that every  $x_i \in X_S$  is output-reachable in the resulting  $\mathcal{G}(\bar{A}, \bar{C})$ .

**Remark 6.** When  $\bar{F} = \bar{I}_n$ , Theorem 4 collapses to the maximum matching based solution to the minimal sensor placement problem for structural observability (cf. Liu et al. 2011).

An example of the procedure of Algorithm 1 is given in Fig. 3. Notably, if  $\bar{A}$  is generically non-diagonalizable, (7) only gives a lower bound for  $\mathcal{P}_1$  (see Fig. 4).

### 7.1.2. General case

In this subsection, we propose two algorithms to obtain an upper bound for  $\mathcal{P}_1$  in the general case, based on a non-decreasing property of SFO w.r.t. a class of edge additions. This bound is proven optimal under certain circumstances.

Given an output matrix  $\bar{C} \in \{0, *\}^{p \times n}$ , denote  $X_C$  as the set of state vertices corresponding to the nonzero columns of  $\bar{C}$  (that is,  $X_C$  is defined similarly to  $X_F$ ), and denote  $E_C = \{(x_i, y_j) : \bar{C}_{ji} \neq 0\}$  as the set of edges corresponding to nonzero entries of  $\bar{C}$ . Recall that  $\mathcal{G}(\bar{A}, \bar{C}) = (X \cup Y, E_{XX} \cup E_{XY})$ . Let  $d(\mathcal{G}(\bar{A}, \bar{C}))$  be the maximum size of an output cactus configuration in  $d(\mathcal{G}(\bar{A}, \bar{C}))$ , i.e., the maximum number of output-reachable state vertices in a collection of disjoint output stems and cycles of  $\mathcal{G}(\bar{A}, \bar{C})$ . Hereafter, “output” will be dropped from “output cactus configuration” for simplicity if no confusion is made. A *maximum cactus configuration* refers to a cactus configuration of  $\mathcal{G}(\bar{A}, \bar{C})$  with the largest size. For a

state vertex  $x_i \in X$  and output vertex  $y_j$ , let  $\mathcal{G}(x_i, y_j)$  be the graph consisting of an edge  $(x_i, y_j)$  from  $x_i$  to  $y_j$  and the corresponding end vertices  $\{x_i, y_j\}$ , i.e.,  $\mathcal{G}(x_i, y_j) = (\{x_i, y_j\}, (x_i, y_j))$ . Given two digraphs  $\mathcal{G}_1 = (V_1, E_1)$  and  $\mathcal{G}_2 = (V_2, E_2)$ ,  $\mathcal{G}_1 \cup \mathcal{G}_2$  denotes the digraph  $(V_1 \cup V_2, E_1 \cup E_2)$ . The following lemma follows directly from Hosoe (1980, Theo 1) and the duality between controllability and observability.

**Lemma 5** (Theo 1, Hosoe, 1980). *Given a structured matrix pair  $(\bar{A}, \bar{C})$ , it holds that  $\text{grank } O(\bar{A}, \bar{C}) = d(\mathcal{G}(\bar{A}, \bar{C}))$ .*

**Lemma 6** (Commault et al., 2017; Iudice et al., 2019). *The triple  $(\bar{A}, \bar{C}, \bar{F})$  is SFO if and only if  $d(\mathcal{G}(\bar{A}, \bar{C})) = d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1}))$  holds for every  $x_i \in X_F$ , i.e., adding an additional dedicated sensor  $y_{p+1}$  to measure state vertex  $x_i$  will not change the generic dimension of unobservable subspaces.*

**Remark 7.** It is worth noting that Lemma 6 has been used in Commault et al. (2017) to find the fixed controllable subspace and Iudice et al. (2019) to characterize the structurally observable nodes of a network, albeit with motivations, notions, or terminologies different from SFO. For self-containedness, it is evident that Lemma 6 follows directly from Lemma 2 (noting that  $\text{grank } [O(\bar{A}, \bar{C}); O(\bar{A}, \bar{F})] = \text{grank } O(\bar{A}, [\bar{C}; \bar{F}])$ ), Corollary 5, and Lemma 5.

**Proposition 4.** *If  $(\bar{A}, \bar{C}, \bar{F})$  is SFO, then  $(\bar{A}, \bar{C}', \bar{F})$  is SFO for any  $\bar{C}'$  such that  $E_C \subseteq E_{C'}$  and  $\text{ini}(\Delta E) \subseteq X_F$ , where  $\Delta E \doteq E_{C'} \setminus E_C$  and  $\text{ini}(\Delta E)$  is the set of initial vertices of edges in  $\Delta E$ .*

**Proof.** The proof is given in the Appendix.  $\square$

Despite the non-monotonicity of SFO w.r.t. edge additions, Proposition 4 reveals that adding edges from functional states to existing sensors would not destroy SFO. We call such a property the *non-decreasing property of SFO w.r.t. functional state measurement addition*. Based on this property, in what follows, we provide two methods to determine an upper bound for  $\mathcal{P}_2$  for general systems, which turns out to be optimal for  $\mathcal{P}_2$  subject to the constraint  $X_C \subseteq X_F$ .

**Algorithm 2 :** A naive iterative algorithm to determine a feasible solution to  $\mathcal{P}_1$  for general systems

- 1: Set  $\bar{C} = \emptyset$
- 2: **while**  $\text{grank } O(\bar{A}, [\bar{C}; \bar{F}]) - \text{grank } O(\bar{A}, \bar{C}) \geq 1$  **do**
- 3:   Update  $\bar{C} = [\bar{C}; \eta]$ , with  $\eta \in \{0, *\}^{1 \times n}$  satisfying  $\eta_i = *$  if and only if  $x_i \in X_F$ .
- 4: **end while**
- 5: Return  $\bar{C}$ .

**Algorithm 3 :** A weighted maximum matching based algorithm for a feasible solution to  $\mathcal{P}_1$  for general systems

- 1: Construct the weighted bipartite graph  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$
- 2: Find the maximum weighted maximum matching  $\mathcal{H}^*$  of  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ . Let  $X_{\mathcal{H}^*} = \{x_i \in X : (x_i, y_j) \in \mathcal{H}^*, y_j \in Y\}$ .
- 3: Construct a  $\gamma \times n$  matrix  $\bar{C}$  such that  $\bar{C}_{ki} = *$  if  $x_i$  is the  $k$ th element (in any order) of  $X_{\mathcal{H}^*}$ ,  $k = 1, \dots, |X_{\mathcal{H}^*}|$ ,  $\gamma \doteq \max\{1, |X_{\mathcal{H}^*}|\}$ .
- 4: **for**  $x_j \in X_F \setminus X_{\mathcal{H}^*}$  **do**
- 5:   **if**  $x_j$  is not reachable to any  $x_k \in X_{\mathcal{H}^*}$  **then**
- 6:     Choose one  $i \in \{1, \dots, \gamma\}$  and let  $\bar{C}_{ij} = *$ ;
- 7:   **end if**
- 8: **end for**
- 9: Return  $\bar{C}$ .

**Naive iterative algorithm:** The first method is a naive iterative algorithm, shown in Algorithm 2. The basic idea is that, since  $\bar{I}_{X_F}$  is a feasible solution to  $\mathcal{P}_1$  (cf. Lemma 2), there exists a  $\bar{C}'$  satisfying  $\text{row}(\bar{C}') \leq |X_F|$  and  $X_{C'} \subseteq X_F$  such that  $(\bar{A}, \bar{C}', \bar{F})$  is SFO. By Proposition 4, whenever there is a feasible solution  $\bar{C}'$  to  $\mathcal{P}_1$  satisfying  $X_{C'} \subseteq X_F$ , the following output matrix

$$\bar{C} = \underbrace{[\eta; \eta; \dots; \eta]}_{\text{row}(\bar{C}') \text{ times}}$$

is also feasible for  $\mathcal{P}_1$  ( $\eta$  is defined in Algorithm 2). Therefore, Algorithm 2 finds a feasible solution  $\bar{C}$  to  $\mathcal{P}_1$  with the minimum number of rows subject to the constraint that  $X_C \subseteq X_F$ .

**Weighted maximum matching based algorithm:** To avoid the iterative steps in Algorithm 2, in what follows, we provide a weighted maximum matching based algorithm to directly determine (i.e., without iterations) the upper bound obtained in Algorithm 2. The key idea is to find the minimum number of output stems in a maximum cactus configuration of  $\mathcal{G}(\bar{A}, \bar{I}_{X_F}) = (X \cup Y, E_{XX} \cup E_{XY})$ , with  $Y = \{y_1, \dots, y_q\}$ ,  $q \doteq |X_F|$ , and  $E_{XX}, E_{XY}$  defined in Section 2. To this end, we construct a weighted bipartite graph  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F}) = (X \cup Y, X \cup Y, \tilde{E}_{XY})$ , where  $\tilde{E}_{XY} = E_{XX} \cup E_{XY} \cup \{(v, v) : v \in X \cup Y\} \cup \{(y, x) : y \in Y, x \in X\}$ . It turns out that  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$  corresponds to a digraph that is obtained from  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$  by adding a self-loop to each vertex of  $X \cup Y$  and a return edge from each vertex of  $Y$  to each one of  $X$  (see Fig. 4(c) for illustration). Let  $W_F \subseteq X$  be the set of state vertices that are output-reachable in  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$ . The edge cost  $w(e) : \tilde{E}_{XY} \rightarrow \mathbb{N}$  is defined as follows:

$$w(e) = \begin{cases} q + 1, & \text{if } e \in E_{XX}^F \\ q, & \text{if } e \in E_{XY} \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where  $E_{XX}^F \doteq \{(x_i, x_j) \in E_{XX} : x_j \in W_F\}$  is the set of state edges within  $W_F$ . Building on  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ , Algorithm 3 finds a feasible solution to  $\mathcal{P}_1$  with the same number of sensors as Algorithm 2 without relying on the iterations; see the following theorem.

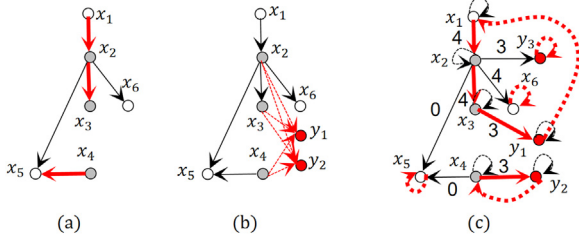
**Theorem 5.** *Both Algorithms 2 and 3 can find a feasible solution to  $\mathcal{P}_1$ , which has the minimum number of rows over all feasible solutions to  $\mathcal{P}_1$  subject to the constraint that  $X_C \subseteq X_F$ . Moreover, if  $\bar{A}$  is generically diagonalizable or  $\bar{F} = \bar{I}_n$ , this solution is optimal for  $\mathcal{P}_1$ .*

**Proof.** Due to its lengthiness, the proof is appended to Appendix.  $\square$

**Example 4.** Consider the system given in Fig. 4(a), where the corresponding  $\bar{A}$  is generically non-diagonalizable. Algorithm 1 applied to this system results in  $p^* = 1$  (see Fig. 4(a) for the maximum matching based interpretation). However, it can be verified via Lemma 2 that for any  $\bar{C} \in \{0, *\}^{1 \times 6}$ ,  $(\bar{A}, \bar{C}, \bar{F})$  is not SFO. Now apply Algorithm 2 to this system. It turns out that if  $\bar{C}_1 = \eta$  with  $\eta = [0, *, *, *, 0, 0]$ , then  $\text{grank } O(\bar{A}, [\bar{C}_1; \bar{F}]) - \text{grank } O(\bar{A}, \bar{C}_1) = 1$ . If  $\bar{C}_2 = [\eta; \eta]$ ,  $\text{grank } O(\bar{A}, [\bar{C}_2; \bar{F}]) - \text{grank } O(\bar{A}, \bar{C}_2) = 0$ . Hence, Algorithm 2 gives a solution  $\bar{C}_2$  to  $\mathcal{P}_1$  with two sensors (see Fig. 4(b) for illustration). Notice that  $(\bar{A}, \bar{C}_2)$  is not structurally observable. Fig. 4(c) presents a set of edges corresponding to a maximum weighted maximum matching in  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ . Associated with it, there are two output stems in  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$ . Therefore, two dedicated sensors deployed on  $x_3$  and  $x_4$  can guarantee the SFO.

**Conjecture:** We conjecture that the solution obtained by Algorithm 3 is optimal for  $\mathcal{P}_1$  even without the constraint  $X_C \subseteq X_F$ . Our extensive examples consistently support this conjecture. However, we are currently unable to provide a formal proof or find a counterexample. Addressing this conjecture remains an open problem for future research.





**Fig. 4.** Illustration of Algorithms 2 and 3 applied to the system digraph  $\mathcal{G}(\bar{A})$  in (a). Nodes in gray represent functional states, and in red represent sensors. In (a), bold red edges form a maximum matching  $\mathcal{M}$  that does not right match  $x_3 \in X_f$  in  $\mathcal{B}(\bar{A})$ , resulting in  $p^* = 1$  via Algorithm 1. In (b), the dotted red edges represent the solution returned by Algorithm 2. Subfigure (c) presents the digraph corresponding to the bipartite graph  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_f})$ , where partial return edges from  $Y$  to  $X$  are omitted for simplicity. The number on each edge is the cost assigned to it via (9), where zero costs are omitted for edges not belonging to  $E_{XX} \cup E_{XY}$ . Bold (solid and dotted) red edges correspond to a maximum weighted maximum matching  $\mathcal{H}^*$  of  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_f})$  with weight  $4 + 4 + 3 + 3 = 14$ . Accordingly,  $X_{\mathcal{H}^*} = \{x_3, x_4\}$ , and there are two output stems  $(x_1, x_2, x_3, y_1)$  and  $(x_4, y_2)$  associated with it in  $\mathcal{G}(\bar{A}, \bar{I}_{X_f})$ . Algorithm 3 then returns the solution of placing two dedicated sensors at  $x_3, x_4$ . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

## 7.2. Minimal actuator placement for SOC

The minimal actuator placement problem seeks to determine the minimum number of actuators (independent inputs), also known as driver nodes (Gao et al., 2014), to achieve SOC. Formally, given  $(\bar{A}, \bar{C})$ , this problem is formulated as

$$\begin{aligned} \min_{\bar{B} \in \{0, * \}^{n \times m}} m \\ \text{s.t. } (\bar{A}, \bar{B}, \bar{C}) \text{ SOC.} \end{aligned} \quad (\mathcal{P}_2)$$

The above problem is denoted by  $\mathcal{P}_2$ . In  $\mathcal{P}_2$ , we do not impose any structure constraint on  $\bar{B}$ . It has been shown in Czeizler et al. (2018, Theo 2) that determining the minimal actuators needed for SOC is NP-hard when the number of states that each actuator can simultaneously actuate is bounded by a constant  $N$ , for any  $N \geq 1$ . The unconstrained  $\mathcal{P}_2$ , however, remains open. In this subsection, we reveal that in the class of generically diagonalizable systems,  $\mathcal{P}_2$  can be solved in polynomial time, in particular, via a weighted maximum flow algorithm.

Given a directed graph (network)  $\mathcal{G} = (V, E)$  with a source node  $s$  and a sink node  $t$ , suppose each edge  $e = (u, v)$  has a capacity  $c(u, v)$  and a cost  $w(u, v)$ . A flow  $f : E \rightarrow \mathbb{R}_{\geq 0}$  over  $\mathcal{G}$  is a map that assigns a non-negative number  $f(e)$  to each edge  $e \in E$  subject to that the flow  $f(e)$  on each edge  $e$  cannot exceed the edge capacity  $c(e)$ , and the sum of flows into a node equals the sum of flows out of that node, unless the source  $s$  and the sink  $t$  (Ahuja et al., 1993). The value of a flow over  $\mathcal{G}$  is the sum of flows passing from the source to the sink, and the cost of a flow  $f$  is defined as  $w(f) = \sum_{e \in E} f(e)w(e)$ . An integral flow refers to a flow  $f$  such that  $f(e)$  is integral for every  $e \in E$ . The maximum flow of a network  $\mathcal{G}$  is the largest possible flow over  $\mathcal{G}$ . A minimum cost maximum flow of  $\mathcal{G}$  is a maximum flow with the smallest possible cost.

**Theorem 6.** Given  $(\bar{A}, \bar{C})$  with  $\bar{A}$  generically diagonalizable and  $\text{grank } \bar{C} = p$ , the optimal value of  $\mathcal{P}_2$  is

$$m^* = \max\{1, p - \text{grank } \bar{C}\bar{A}\},$$

and Algorithm 4 can determine the corresponding optimal solution  $\bar{B}^*$  in  $O(n^3)$  time.

**Proof.** Suppose  $(\bar{A}, \bar{B}, \bar{C})$  is SOC. From Corollary 4 and the proof of Theorem 3, it holds that  $\text{grank } \bar{C}[\bar{A}_r, \bar{B}] = p$ . Noting  $\text{grank } \bar{C}[\bar{A}_r, \bar{B}] \leq \text{grank } \bar{C}[\bar{A}, \bar{B}] \leq \text{grank } \bar{C}\bar{A} + \text{grank } \bar{C}\bar{B}$ , we have

$\text{row}(\bar{B}) \geq \text{grank } \bar{C}\bar{B} \geq p - \text{grank } \bar{C}\bar{A}$ . This indicates  $m^*$  is a lower bound for  $\mathcal{P}_2$ .

By the construction in Algorithm 4, the maximum flow of the network  $\mathcal{F}(\bar{A}, \bar{C})$  is exactly  $p$ . From the integral flow theorem (Ahuja et al., 1993), the minimum cost integral maximum flow  $f^*$  exists. Moreover, all integral maximum flows are of one-one correspondence to linkings with size  $p$  from  $U^2 \cup X^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}, \bar{I}_n, \bar{C})$ . Since  $\text{grank } \bar{C}\bar{A}$  equals the maximum size of a linking from  $X^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}, \bar{I}_n, \bar{C})$  (Papadimitriou and Tsitsiklis 1984, Theo. 6), we have  $|X^{f^*}| = \text{grank } \bar{C}\bar{A}$  and  $|X^{f^*}| = p - \text{grank } \bar{C}\bar{A}$ . In addition, Step 3 of Algorithm 4 ensures that the initial vertex of each path in the linking of size  $|X^{f^*}|$  from  $X^2$  to  $Y^0$  in  $\mathcal{D}(\bar{A}, \bar{B}^*, \bar{C})$  is input-reachable. Therefore,  $(\bar{A}, \bar{B}^*, \bar{C})$  is SOC, and  $\bar{B}^*$  has the minimum number  $m^*$  of columns. The  $O(n^3)$  complexity comes from the fact that a minimum cost maximum flow of a network  $\mathcal{G} = (V, E)$  can be determined in  $O(|V||E|)$  time (Ahuja et al., 1993).  $\square$

**Algorithm 4 :** A weighted maximum flow based algorithm to find an optimal solution to  $\mathcal{P}_2$  in generically diagonalizable systems

**Require:**  $\bar{A}$  and  $\bar{C}$  with  $\bar{A}$  generically diagonalizable and  $\text{grank } \bar{C} = p$ .

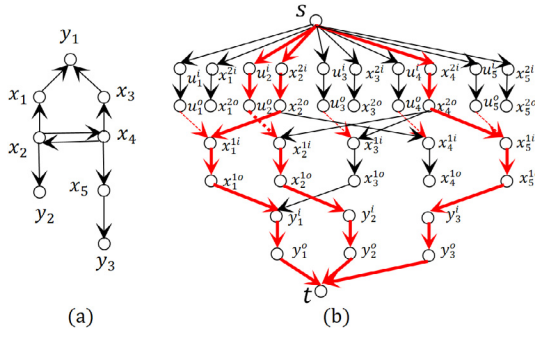
**Ensure:** An optimal solution  $\bar{B}^*$  to  $\mathcal{P}_2$ .

- 1: Construct a flow network  $\mathcal{F}(\bar{A}, \bar{C})$  from  $\mathcal{D}(\bar{A}, \bar{I}_n, \bar{C})$  as follows: duplicate each vertex  $v$  of  $\mathcal{D}(\bar{A}, \bar{I}_n, \bar{C})$  with two vertices  $v^i, v^o$  and an edge  $(v^i, v^o)$ , and for each edge  $(v, w)$  of  $\mathcal{D}(\bar{A}, \bar{I}_n, \bar{C})$ , replace it with an edge  $(v^o, w^i)$ ; add to the resultant graph a source  $s$ , a sink  $t$ , and the incident edges  $\{(s, v^i) : v \in U^2 \cup X^2\} \cup \{(w^o, t) : w \in Y^0\}$ . The edge capacity is set as  $c(e) = 1$  for each edge  $e$  of  $\mathcal{F}(\bar{A}, \bar{C})$ . The edge cost is set as  $w(e) = 1$  if  $e \in \{(u_j^o, x_j^{i1}) : j = 1, \dots, n\}$ , and  $w(e) = 0$  otherwise.
- 2: Find a minimum cost integral maximum flow  $f^*$  of  $\mathcal{F}(\bar{A}, \bar{C})$ . Let  $X^{f^*}_1 = \{x_i \in X : f(u_i^o, x_i^{i1}) \neq 0\}$ ,  $X^{f^*}_2 = \{x_i \in X : f(x_i^{i2}, x_i^{i2o}) \neq 0\}$ , and  $m^* = \max\{1, |X^{f^*}_2|\}$ . Suppose that  $X^{f^*}_2$  are located at  $q$  different SCCs of  $\mathcal{G}(\bar{A})$ , with the vertex set of the  $i$ th one being  $X_i \subseteq X$ .
- 3: Construct an  $n \times m^*$  matrix  $\bar{B}^*$  such that  $\bar{B}^*_{jk} = *$  if  $x_j$  is the  $k$ th element of  $X^{f^*}_1$ ,  $k = 1, \dots, |X^{f^*}_1|$ . In addition, per  $i \in \{1, \dots, q\}$ , choosing one  $x_j \in X_i$  and one  $k \in \{1, \dots, m^*\}$ , let  $\bar{B}^*_{jk} = *$ .
- 4: Return  $\bar{B}^*$ .

**Example 5.** Consider a system  $(\bar{A}, \bar{C})$  whose  $\mathcal{G}(\bar{A}, \bar{C})$  is given in Fig. 5(a). It can be verified that  $\bar{A}$  is generically diagonalizable. Fig. 5(b) presents the corresponding flow network  $\mathcal{F}(\bar{A}, \bar{C})$  constructed in Algorithm 4. In the minimum cost maximum flow  $f^*$  (edges with nonzero flows are highlighted in bold in Fig. 5(b)),  $X^{f^*}_1 = \{x_2\}$  and  $X^{f^*}_2 = \{x_2, x_4\}$ , yielding  $m^* = 1$ . From step 3 of Algorithm 4, deploying a dedicated actuator at  $x_2$ , corresponding to  $\bar{B}^* = [0, *, 0, 0, 0]^T$ , suffices to make the resulting system SOC. Another optimal solution is deploying a non-dedicated actuator that connects to  $x_5$  and  $x_4$  (or  $x_5$  and  $x_2$ ) simultaneously.

**Remark 8.** Since adding additional links from actuators to states would not destroy SOC, if SOC can be verified in polynomial time, then a naive iterative algorithm like Algorithm 2 can find an optimal solution to  $\mathcal{P}_2$  for general systems. By contrast, even though SFO can be verified in polynomial time, the complexity of  $\mathcal{P}_1$  remains open.

**Remark 9.** The Matlab codes for Algorithms 1–4, as well as for checking generic diagonalizability (Proposition 2), are available at <https://github.com/Yuanzhang2014/Generic-diagonalizability-SFO-SOC>.



**Fig. 5.** Illustration of Algorithm 4. Figure (a) gives the digraph  $\mathcal{G}(\bar{A}, \bar{C})$ . Figure (b) depicts the corresponding flow network  $\mathcal{F}(\bar{A}, \bar{C})$  constructed in step 1 of Algorithm 4. All edges have capacity one, dotted red edges have cost one, and the rest have cost zero. Bold red edges form a minimum cost maximum flow  $f^*$  with total cost one, where each edge has flow 1. Associated with it,  $X^{f^*}_1 = \{x_2\}$  and  $X^{f^*}_2 = \{x_2, x_4\}$ . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

## 8. Conclusions

This paper has provided polynomial-time solutions to the verification and optimal sensor/actuator placement problems associated with SFO and SOC in a specific class of systems, namely, generically diagonalizable systems. We have defined generically diagonalizable matrices and provided their graph-theoretic characterizations. Additionally, we have derived computationally efficient criteria for assessing SFO and SOC in such systems. For these systems, we have provided closed-form solutions, as well as graph-theoretic algorithms to determine the minimal sensors to achieve SFO and the minimal actuators to achieve SOC, respectively. For more general systems to achieve SFO, we have established an upper bound by identifying a non-decreasing property of SFO w.r.t. a specific class of edge additions, which is proven to be optimal under certain conditions.

Our work has highlighted that structural diagonalizability can significantly simplify problems related to SFO and SOC. We conjecture that the ratio of structurally diagonalizable networks within the set of all networks with  $n$  nodes approaches one as  $n$  approaches infinity. Future work includes proving this conjecture, exploring additional properties of generically diagonalizable systems, and investigating the complexity of the sensor placement problem  $\mathcal{P}_1$  in the general case.

## Acknowledgment

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## Appendix

**Proof of Proposition 4.** It suffices to consider the case where  $\text{row}(\bar{C}) = \text{row}(\bar{C}') = p$ , since adding more nonzero rows (i.e. sensors) to  $\bar{C}'$  would not destroy the SFO provided that  $(\bar{A}, \bar{C}', \bar{F})$  is SFO. Without loss of generality, suppose  $|\Delta E| \geq 1$ . Let  $e \in \Delta E$  be an arbitrary edge of  $\Delta E$ . Consider  $e = (x_i, y_j)$ , where  $x_i \in X_F$  and  $y_j \in Y \doteq \{y_1, \dots, y_p\}$ . Let  $\bar{C}^e$  be the output matrix obtained from  $\bar{C}$  after adding  $e$  to  $E_C$ . We first show that  $d(\mathcal{G}(\bar{A}, \bar{C}^e)) = d(\mathcal{G}(\bar{A}, \bar{C}))$  by contradiction. If  $d(\mathcal{G}(\bar{A}, \bar{C}^e)) > d(\mathcal{G}(\bar{A}, \bar{C}))$ , let  $\mathcal{G}_{\text{cact}}$  be an arbitrary maximum cactus configuration in  $\mathcal{G}(\bar{A}, \bar{C}^e)$ . Since  $(\bar{A}, \bar{C}, \bar{F})$  is SFO, by Zhang et al. (2025b, Coro 4), every  $x \in X_F$  is output-reachable in  $\mathcal{G}(\bar{A}, \bar{C})$ , which means adding  $e$  to  $\mathcal{G}(\bar{A}, \bar{C})$  will not affect the output-reachability of state vertices of  $\mathcal{G}(\bar{A}, \bar{C})$ . As

a result,  $e$  must be contained in  $\mathcal{G}_{\text{cact}}$ . Observe that by changing  $e$  in  $\mathcal{G}_{\text{cact}}$  to  $(x_i, y_{p+1})$ , we can obtain a new cactus configuration of size  $d(\mathcal{G}(\bar{A}, \bar{C}^e))$  in  $\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1})$ . Consequently, it holds that

$$d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1})) \geq d(\mathcal{G}(\bar{A}, \bar{C}^e)). \quad (10)$$

Inequality (10) and the assumption  $d(\mathcal{G}(\bar{A}, \bar{C}^e)) > d(\mathcal{G}(\bar{A}, \bar{C}))$  yield  $d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1})) > d(\mathcal{G}(\bar{A}, \bar{C}))$ , which contradicts the SFO of  $(\bar{A}, \bar{C}, \bar{F})$  from Lemma 6. Therefore, it holds that  $d(\mathcal{G}(\bar{A}, \bar{C}^e)) = d(\mathcal{G}(\bar{A}, \bar{C}))$ .

We now prove that  $(\bar{A}, \bar{C}^e, \bar{F})$  is SFO by showing the SFO of  $(\bar{A}, \bar{C}^e, \bar{e}_{i^*})$  for each  $x_{i^*} \in X_F$ . First, consider the case  $i^* = i$ . If  $(\bar{A}, \bar{C}^e, \bar{e}_i)$  is not SFO, Lemma 6 yields  $d(\mathcal{G}(\bar{A}, \bar{C}^e) \cup \mathcal{G}(x_i, y_{p+1})) > d(\mathcal{G}(\bar{A}, \bar{C}^e))$ . However, by the similar reasoning to (10), it holds

$$d(\mathcal{G}(\bar{A}, \bar{C}^e) \cup \mathcal{G}(x_i, y_{p+1})) = d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1})), \quad (11)$$

which comes from the fact that every state vertex must have a unique outgoing edge in a cactus configuration of  $\mathcal{G}(\bar{A}, \bar{C}^e) \cup \mathcal{G}(x_i, y_{p+1})$ , and each state vertex  $x \in X_F$  has already been output-reachable in  $\mathcal{G}(\bar{A}, \bar{C})$ . These relations together yield  $d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1})) > d(\mathcal{G}(\bar{A}, \bar{C}))$ , contradicting the SFO of  $(\bar{A}, \bar{C}, \bar{e}_i)$ . Therefore,  $(\bar{A}, \bar{C}^e, \bar{e}_i)$  is SFO. Next, consider the case  $i^* \neq i$ . Let  $y_{p+2}$  be the  $(p+2)$ th output vertex. By the similar reasoning to (10) and (11), we have  $d(\mathcal{G}(\bar{A}, \bar{C}^e) \cup \mathcal{G}(x_{i^*}, y_{p+2})) \leq d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1}) \cup \mathcal{G}(x_{i^*}, y_{p+2}))$ . On the other hand, from condition (d) of Corollary 5, the SFO of  $(\bar{A}, \bar{C}, \bar{F})$  requires that  $(\bar{A}, \bar{C}, \bar{F}')$  is SFO with  $X_{F'} = \{x_i, x_{i^*}\}$ . This fact with Lemma 6 yields  $d(\mathcal{G}(\bar{A}, \bar{C}) \cup \mathcal{G}(x_i, y_{p+1}) \cup \mathcal{G}(x_{i^*}, y_{p+2})) = d(\mathcal{G}(\bar{A}, \bar{C}))$ . Since  $d(\mathcal{G}(\bar{A}, \bar{C}^e) \cup \mathcal{G}(x_{i^*}, y_{p+2})) \geq d(\mathcal{G}(\bar{A}, \bar{C}^e))$ , we reach  $d(\mathcal{G}(\bar{A}, \bar{C}^e) \cup \mathcal{G}(x_{i^*}, y_{p+2})) = d(\mathcal{G}(\bar{A}, \bar{C}))$ , which implies that  $(\bar{A}, \bar{C}^e, \bar{e}_{i^*})$  is SFO. Since this holds for each  $x_{i^*} \in X_F$ ,  $(\bar{A}, \bar{C}^e, \bar{F})$  is SFO via Corollary 5.

Let  $\bar{C}^{e'}$  be the output matrix corresponding to  $\{e'\} \cup E_{\bar{C}^e}$  for any  $e' \in \Delta E \setminus \{e\}$ . Similarly to the above analysis, we can prove that  $(\bar{A}, \bar{C}^{e'}, \bar{F})$  is SFO from the SFO of  $(\bar{A}, \bar{C}^e, \bar{F})$ . Repeat this procedure  $|\Delta E|$  times, and we obtain that  $(\bar{A}, \bar{C}', \bar{F})$  is SFO.  $\square$

**Proof of Theorem 5.** The feasibility of Algorithm 2 follows from the non-decreasing property of SFO w.r.t. functional state measurement addition as analyzed above. Below, we focus on the statements about Algorithm 3. First, observe that given a collection of disjoint cycles and stems in  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$  (note that vertices in the cycles need not be output-reachable), we can complete each stem to a cycle by adding the zero cost edge from its terminal vertex in  $Y$  to its initial vertex in  $X$ , and then add the loops for uncovered vertices in  $X \cup Y$ . Then  $X \cup Y$  are covered by disjoint cycles, corresponding to a maximum (perfect) matching of  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ . Conversely, given a maximum matching in  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ , by omitting the zero cost edges, we obtain a collection of disjoint cycles and stems (as well as isolated vertices) in  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$ . Consider a maximum matching  $\mathcal{H}$  of  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$  and let the indicator function  $\mathbb{I} : \tilde{E}_{XY} \rightarrow \{0, 1\}$  be such that  $\mathbb{I}(\mathcal{H}, e) = 1$  if  $e \in \mathcal{H}$  and  $\mathbb{I}(\mathcal{H}, e) = 0$  otherwise. Observe that the weight  $w(\mathcal{H})$  of a maximum matching  $\mathcal{H}$  in  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$  can be equivalently written as

$$w(\mathcal{H}) \doteq (q+1) \sum_{e \in E_{XY}^F \doteq E_{XX}^F \cup E_{XY}} \mathbb{I}(\mathcal{H}, e) - \sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}, e).$$

Suppose  $\mathcal{H}^*$  is a maximum matching with the maximum weight in  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ . Then, it must hold  $\sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}^*, e) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$ . To show this, assume, for the sake of contradiction,  $\sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}^*, e) < \text{grank } O(\bar{A}, \bar{I}_{X_F})$  (noting that all vertices in the set  $X \setminus W_F$  are output-unreachable, by Lemma 5, we have  $\sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}, e) \leq d(\mathcal{G}(\bar{A}, \bar{I}_{X_F})) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$  for any maximum matching  $\mathcal{H}$  of  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$ ). Consider a maximum matching  $\mathcal{H}'$  in  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$  such that  $\sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}', e) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$  (such a maximum matching always exists by Lemma 5). Observe

that

$$\begin{aligned} w(\mathcal{H}^*) - w(\mathcal{H}') &= (q+1) \left( \sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}^*, e) \right. \\ &\quad \left. - \sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}', e) \right) - \left( \sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}^*, e) - \sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}', e) \right) \\ &\leq -(q+1) + q \\ &< 0, \end{aligned}$$

where the first inequality results from the fact that  $0 \leq \sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}', e)$ ,  $\sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}^*, e) \leq q$ . This contradicts the assumption that  $\mathcal{H}^*$  is a maximum weighted maximum matching. As a result of the above analysis,  $\mathcal{H}^*$  has the minimum  $\sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}^*, e)$  over all perfect matchings  $\mathcal{H}$  in  $\tilde{\mathcal{B}}(\bar{A}, \bar{I}_{X_F})$  satisfying  $\sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}, e) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$ . Equivalently,  $\mathcal{H}^*$  corresponds to a maximum cactus configuration of  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$  with the minimum number of output stems (this number equals  $\sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}^*, e)$ ). It turns out that  $\gamma \doteq \max\{1, \sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}^*, e)\}$  is the minimum number of rows of an output matrix  $\bar{C}$  subject to  $X_C \subseteq X_F$  such that  $\text{grank } O(\bar{A}, \bar{C}) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$ . Indeed, if there is another  $\bar{C}'$  with fewer rows than  $\bar{C}$  that satisfies the aforementioned condition, then a maximum cactus configuration in  $\mathcal{G}(\bar{A}, \bar{C}')$  exists with fewer output stems than  $\sum_{e \in E_{XY}} \mathbb{I}(\mathcal{H}^*, e)$ , which is also a maximum cactus configuration in  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$ , causing a contradiction. Now consider the output matrix  $\bar{C}$  found by Algorithm 3. By the construction, any state vertex that is output-reachable in  $\mathcal{G}(\bar{A}, \bar{I}_{X_F})$  is also output-reachable in  $\mathcal{G}(\bar{A}, \bar{C})$ . Consequently,  $\text{grank } O(\bar{A}, \bar{C}) = \sum_{e \in E_{XY}^F} \mathbb{I}(\mathcal{H}^*, e) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$ . Since  $X_C \subseteq X_F$ , it holds that  $\text{grank } O(\bar{A}, [\bar{C}; \bar{I}_{X_F}]) = \text{grank } O(\bar{A}, \bar{I}_{X_F})$ , leading to  $\text{grank } O(\bar{A}, \bar{C}) = \text{grank } O(\bar{A}, [\bar{C}; \bar{I}_{X_F}])$ . By Corollary 5,  $(\bar{A}, \bar{C}, F)$  is SFO.

The optimality when  $\bar{F} = \bar{I}_n$  is obvious from the above analysis. Since the optimal solution  $\bar{C}$  returned by Algorithm 1 satisfies  $X_C \subseteq X_F$ , the solution given by Algorithm 2 or 3 is also optimal for generically diagonalizable systems.  $\square$

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